

REFLECTION SUBGROUPS AND THE HOMOTOPY THEORIES OF CUBICAL SITES

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ABSTRACT. Every standard cube category — cartesian, Dedekind, De Morgan, Kleene, Boolean — carries two canonical homotopy theories on its presheaves: the cubical-type model structure of cubical type theory and Cisinski’s test structure, which presents spaces. This paper determines, uniformly in the site and in the dimension, how the two theories interact on the natural finite testing ground of isotropy quotients $\square^n/H$, and shows that the entire interaction is controlled by reflection subgroups. Three layers of results. *The phase diagram:* on sites with connections and no reversal, every isotropy quotient is contractible in both structures, so no quotient witness can exist — while on each of the three reversal sites the explicit map-level and object-level separations of the De Morgan case transfer uniformly, because the proofs’ site-dependence is concentrated in two evaluations at unmixed points. *The landscape:* across the reversal sites the classification’s conclusions coincide but its proofs cannot — a rigidity discriminant shows that twisted invariant cells slide precisely when a strata-swap character vanishes, genuinely twisted cells survive every stage of fibrant replacement (detected by test-side covering theory), and their existence is governed by character realizability in the free algebras, where the De Morgan site is the exceptional one. A covering theory for uniform Kan replacements — coverings extend uniquely along the algebraic fibrant replacement — yields a descent theorem for weak equivalences along free quotients. *The criterion:* for a mixed subgroup H , let N be the normal closure of its reflections. The quotient $\square^n/H$ agrees if N still fixes a cell — an unsigned retraction contracts $\square^n/N$ and the reflection-free group H/N acts freely, so both structures see $B(H/N)$ — and carries an attached separation if N is mixed, *in every dimension:* on the Boolean site a sieve-landing dichotomy shows that the strict homotopy component of the identity of $\square^n/N$ consists exactly of twisted-equivariant vertex maps, so either none is stabilized and the attached witness separates outright, or one is and the quotient is equivalent, in both structures simultaneously, to a strictly smaller instance of the same problem — a quotient of a codiscrete object by a fixed-point-free image group that a transport lemma keeps generated by fixed-point-having elements. The recursion terminates, and the criterion becomes a theorem at every n : in dimensions ≤ 3 it reproduces the known classification of all 98 subgroups with zero mismatches, and in dimension 4 a closed-form census decides all 30 reflection-generated mixed classes (28 rigid, 2 reductions — one landing on the Klein quotient itself, the other on an order-4 mixed block in dimension three). A byproduct is a bridge to the homotopy theory of finite group actions: the orbit space of the Milnor construction of a finite M -set is $\mathbb{F}_2$ -acyclic whenever a Sylow 2-subgroup has a fixed point. All finite claims are machine-verified in a public, version-pinned artifact.

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1. INTRODUCTION

Cubical type theories interpret dependent type theory in presheaves over a cube category, and the choice of cube category is the choice of an algebraic theory of the interval: faces and degeneracies only (the cartesian site of [1]), binary connections (the Dedekind site), a De Morgan involution (the site of CCHM type theory [6]), the Kleene inequality, or full Boolean complementation. Each site carries the *cubical-type model structure*, whose fibrations are the uniform Kan fibrations of the type-theoretic interpretation [6, 11, 15], and the *test model structure* of Cisinski [8], which presents the homotopy theory of spaces; the containment $\mathbf{W}_{\text{type}} \subseteq \mathbf{W}_{\text{test}}$ holds on all of them. Whether it is an equality — whether cubical type theory *presents spaces* — was expected to fail on the sites with reversal since a 2018 folklore argument of Sattler and Coquand, and the expectation was recently confirmed on the De Morgan site in a strong form: there is an explicit separating map, and an explicit finite presheaf that is contractible in the test structure but not in the type structure [16]; in dimensions $n \leq 3$, the isotropy quotients $\square^n/H$, $H \leq B_n$ the hyperoctahedral group, were completely classified into 78 agreement and 20 separation cases [9].

This paper is about what happens *beyond* the De Morgan site and *beyond* the case analysis. Its unifying discovery is that a single algebraic actor controls the entire interaction of the two homotopy theories with isotropy:

the reflection subgroup — the subgroup generated by the elements that fix a cell.

The paper’s three layers make this precise at increasing depth.

Layer 1: the phase diagram. Which interval theories admit isotropy phenomena at all?

Theorem 1.1 (Phase diagram; Part 1). *(i) On every cubical site whose interval theory contains a connection and no reversal, every isotropy quotient $\square^n/H$ is contractible in both structures: a single equivariant connection contraction kills all witnesses. In particular the openness of the Dedekind comparison is structural: no quotient witness can decide it. (ii) On each of the De Morgan, Kleene, and Boolean sites, the explicit map-level and object-level separations of [16] hold verbatim: the separating median collage and the test-contractible, non-type-contractible subdivided cone exist on all three. The transfer is universal-algebraic: all term identities descend to subvarieties, and the only site-specific inputs are two parity lemmas whose witness points are unmixed.*

Layer 2: the landscape. On the three reversal sites the classification’s conclusions coincide — but, provably, its proofs cannot.

Theorem 1.2 (Landscape; Part 2). *(i) (Rigidity discriminant) An invariant interior cell of a witness machine with twist δ is stage-zero slide-reachable to an end iff the character $\varepsilon \circ \delta$ vanishes, where ε is the strata-swap character; genuinely twisted cells survive every stage of the fibrant replacement, detected by a strict character projection to the interval quotient — test-side $\mathbb{Z}/2$ -covering theory obstructing type-side nullity. (ii) (Realizability divide) Twisted cells exist over a site iff the character is realizable by a twisted free-algebra element: always on the Boolean site (signed XOR), always in the relevant cases on the Kleene site (the Kleene inequality removes precisely the mixed points that carry the De Morgan parity obstructions), and essentially never on the De Morgan site — the site of cubical type theory is the exceptional site, the one where the certificate method happens to work. (iii) (Covering descent) Presheaf coverings are local systems on the category of elements and extend uniquely along the algebraic fibrant replacement; free quotients admit equivariant replacements, and a map of free quotients is a type weak equivalence iff its equivariant cover is. Separations therefore descend along towers, bypassing the broken certificates.*

Layer 3: the criterion. The case analysis of [9] — four agreement families, three separation mechanisms — collapses into one statement.

Theorem 1.3 (Criterion; Part 3). *Let $H \leq B_n$ be mixed, on any reversal site, and let $N := \langle\langle \text{Refl}(H) \rangle\rangle$ be the normal closure of its reflections.*

- (1) *If N has a common fixed cell, then $\square^n/H \simeq B(H/N)$ in both structures.*
- (2) *If N is mixed, an explicit test weak equivalence that is not a type weak equivalence is attached to H : on the Boolean site for every n (Theorem 15.8, via the sieve-landing dichotomy and the dimension-reduction recursion), and on the De Morgan and Kleene sites for $n \leq 3$ by the landscape transfer of Part 2.*

The engine of (2) is an exact characterization: the obstruction-certificate method succeeds for a group if and only if the group is generated by its reflections — because certificate consistency forces every twist to kill all vertex stabilizers, nontrivial vertex stabilizers consist of reflections, and every reflection fixes a vertex; N is reflection-generated by construction, and the separation descends along the free cover H/N by (iii) above.

In dimension ≤ 3 the criterion reproduces the 78/20 classification with zero mismatches — on the De Morgan site by direct comparison with [9], on the Kleene and Boolean sites via the landscape transfer; in dimension 4 the closed-form census decides all 30 reflection-generated mixed classes — 28 rigid, where the attached witness separates, and 2 tree-arrangement classes where the quotient reduces, in both structures at once, to the Klein quotient and to an order-4 mixed block. The agreement half of the criterion is itself a simplification discovery: the median retraction of [9], whose odd-arity self-dual majority term had seemed essential, is an avoidable device — retracting at the level of N , where no signs act, and only then passing to the free cover eliminates the equivariance problem the median was solving, and simultaneously handles degenerate groups (found by machine search in dimension 4) for which *every* one-step equivariant retraction provably fails.

Methods and verification. The paper is written against a public, version-pinned artifact: every finite combinatorial claim — censuses, certificates, discriminants, towers, the 98-subgroup reproduction — is verified by exhaustive computation, with exact (parity- and satisfiability-based) decision procedures on the Boolean and Kleene sites. Several of the main theorems were found as empirical laws in this data — the rigidity discriminant, the realizability divide, the criterion itself — and then proved.

Related work. The comparison problem originates in the type-theoretic literature: the cartesian equivariant repair of Awodey, Cavallo, Coquand, Riehl and Sattler [1], the elimination of reversals from cubical type theories by Cavallo and Sattler [7], and the model-structure frameworks of [11, 15]. The explicit De Morgan separation and the three-dimensional classification are in the companion papers [16, 9]; the present paper subsumes and extends the archived series on the phase diagram and the landscape. The test-category side rests on [8, 13, 5]; free Kleene algebras on [3]. The slogan that reflection subgroups control the comparison appears to be new; its resonance with the theory of reflection arrangements is discussed in the outlook. The

fundamental-group formula for isotropy quotients (Theorem 14.6) is a categorical relative of Armstrong's theorem on quotients of simply-connected spaces [2]; the codiscrete objects $E(T)$ are Milnor constructions [12] and, equivariantly, classifying spaces for the family of subgroups with fixed points, so the arrangement dichotomy of the outlook sits beside Quillen's poset conjecture [14] and the orbit-space theorems of Webb and Symonds [19, 17].

Part 1. The phase diagram

2. CUBICAL SITES FROM VARIETIES

2.1. The sites. Let $\mathcal{V}$ be a variety of algebras on the signature $\{\wedge, \vee, 0, 1\}$ or $\{\wedge, \vee, \neg, 0, 1\}$ containing the bounded distributive lattice axioms, and write $F_{\mathcal{V}}(n)$ for the free algebra on $x_1, \dots, x_n$. The *cube category* $\square_{\mathcal{V}}$ has objects $[n]$ for $n \geq 0$ and morphisms

$$\square_{\mathcal{V}}([m], [n]) = \text{Hom}_{\mathcal{V}}(F_{\mathcal{V}}(n), F_{\mathcal{V}}(m)) \cong F_{\mathcal{V}}(m)^n,$$

composition being substitution; presheaves on $\square_{\mathcal{V}}$ are $\mathcal{V}$ -cubical sets. We consider:

- the *Dedekind family*: $\mathcal{V} \supseteq$ bounded distributive lattices, no $\neg$ (this includes the Dedekind site itself, $F_{\mathcal{V}}(n)$ the free bounded distributive lattice, of cardinality the Dedekind numbers 3, 6, 20, 168, $\dots$);
- the *De Morgan site* $\square_{\text{DM}}$: $\mathcal{V} =$ De Morgan algebras (distributive lattices with an involution $\neg$ satisfying the De Morgan laws); $|F_{\text{DM}}(2)| = 168$;
- the *Kleene site* $\square_{\text{KL}}$: the subvariety of De Morgan algebras with $x \wedge \neg x \leq y \vee \neg y$; $|F_{\text{KL}}(n)| = 6, 84, 43918, \dots$ [3];
- the *Boolean site* $\square_{\text{BA}}$: the subvariety of Boolean algebras; $|F_{\text{BA}}(n)| = 2^{2^n}$;
- the *cartesian site* $\square_{\text{cart}}$: no lattice operations; morphisms are tuples of variables and constants.

All of these are strict test categories: the cube category with connections is a strict test category by Maltsiniotis [13], and the remaining cases follow from the criteria of [8]; see the systematic treatment of Buchholtz and Morehouse [5]. Hence each site carries the test model structure, with $\mathbf{W}_{\text{test}}$ created by the nerve of the category of elements $\text{el}(-)$, and each carries the cubical-type model structure with cofibrations the monomorphisms and fibrations the uniform Kan fibrations generated by the open-box inclusions $(m \hat{\otimes} \delta^{\varepsilon})$ [11, 15]. The containment (U) holds on every strict test cube category by the two-out-of-three argument of [16, Prop. 2.5] (which we do not repeat): the generating trivial cofibrations are monic test equivalences, and type-trivial fibrations have the right lifting property against all monomorphisms.

2.2. Symmetries and isotropy quotients. The invertible morphisms $[n] \rightarrow [n]$ of $\square_{\mathcal{V}}$ form the symmetric group Σ_n when $\neg \notin \mathcal{V}$ and the signed permutation (hyperoctahedral) group $B_n = \Sigma_n \ltimes (\mathbb{Z}/2)^n$ when $\neg \in \mathcal{V}$. For

H a subgroup of this automorphism group, the *isotropy quotient* $\square^n/H$ is the levelwise orbit presheaf. These quotients are the natural finite testing ground for the comparison: they are where all known object-level phenomena live [18, 9, 16].

Definition 2.1. An element of B_n is a *reflection* if it fixes a nonempty cell of $\square^n$ — equivalently (Lemma 11.1), if every cycle of its signed cycle decomposition has even sign sum, equivalently if it fixes a vertex. A subgroup $H \leq B_n$ is *mixed* if it fixes no nonempty cell (for the group, fixing a cell and fixing a vertex again agree). We write $\text{SEP}(H)$ for the statement that an explicit map in $\mathbf{W}_{\text{test}} \setminus \mathbf{W}_{\text{type}}$ is attached to H — built naturally from $\square^n/H$ -data, as in the criterion — and call H an *agreement* class when $\square^n/H$ has equivalent homotopy types in the two structures via a common B -object.

2.3. Free-algebra representations. Computations use the standard function representations. Free De Morgan algebras are monotone $\{0, 1\}$ -valued functions on the literal cube $L_n = \{0, 1\}^{2n}$ (coordinates in pairs $(x_i, \neg x_i)$), with negation $(\neg\varphi)(p) = 1 - \varphi(\rho p)$, where ρ swaps and complements each pair. Free Kleene algebras are the monotone functions on the *unmixed* subposet $U_n \subseteq L_n$ — the points having no $(0, 0)$ -pair together with a $(1, 1)$ -pair — with the same formula for negation (ρ preserves U_n) [3]. Free Boolean algebras are all $\{0, 1\}$ -valued functions on $\{0, 1\}^n$, with negation the output complement. A point of L_n is called unmixed when it lies in U_n ; the transfer of the parity lemmas in Section 4 rests on the observation that the De Morgan proofs only ever evaluate at unmixed points.

3. CONNECTIONS KILL ISOTROPY WITNESSES

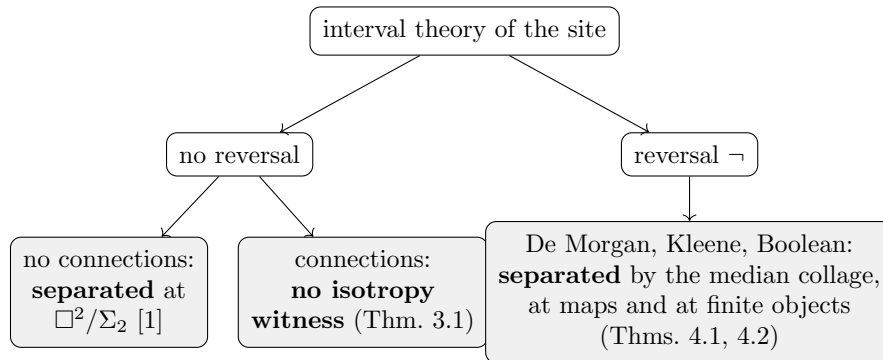


FIGURE 1. The phase diagram of cubical sites.

Theorem 3.1 (Connections kill isotropy witnesses). *Let $\square_{\mathcal{V}}$ be a cube category whose interval theory contains a connection ($\vee$ or $\wedge$) and no reversal, so that the automorphisms of $[n]$ in $\square_{\mathcal{V}}$ form the symmetric group Σ_n . Then*

for every n and every $H \leq \Sigma_n$ the isotropy quotient $\square^n/H$ admits a strict H -equivariant contraction, and is therefore contractible in both the type and the test structure. In particular the two structures agree on all isotropy quotients of $\square_{\mathcal{V}}$, and no analogue of the known object-level witnesses exists on such sites.

Proof of Theorem 3.1. Suppose $\vee \in \mathcal{V}$ (the case of $\wedge$ is dual, using the 0-endpoint). Consider the cell

$$c = (x_1 \vee t, \dots, x_n \vee t) \in \square_{\mathcal{V}}([n+1], [n]),$$

regarded as a map $c : \square^n \otimes \square^1 \rightarrow \square^n$ of representables, where t is the cylinder coordinate. Then: (i) $c|_{t=0} = \text{id}$ and $c|_{t=1}$ is the constant map at the vertex $(1, \dots, 1)$; (ii) c is Σ_n -equivariant: permuting the x_i commutes with the coordinatewise formula, and the vertex $(1, \dots, 1)$ is fixed; (iii) since quotients of presheaves are levelwise and the cylinder is formed in the x -coordinates only, c descends to a homotopy

$$\bar{c} : (\square^n/H) \times \square^1 \longrightarrow \square^n/H, \quad \bar{c}_0 = \text{id}, \quad \bar{c}_1 = \text{const},$$

for every $H \leq \Sigma_n$. A strict homotopy between the identity and a constant is a left homotopy in any model structure in which the projection $X \times \square^1 \rightarrow X$ is a weak equivalence, which holds in the type structure (the interval is contractible) and in the test structure (the site is a strict test category). Hence $\square^n/H \rightarrow 1$ is a weak equivalence in both structures. $\square$

Corollary 3.2. *On the Dedekind site the type and test structures agree on all isotropy quotients. In particular, whether the Dedekind site presents spaces cannot be decided by an isotropy witness.*

Remark 3.3. On the cartesian site the contraction of Theorem 3.1 is unavailable — there is no cell $x \vee t$ — and indeed the symmetric square $Q = \square^2/\Sigma_2$ is a witness: it is test-contractible, while its failure to be type-contractible is the observation of Coquand and Sattler that motivated the equivariant model structure of [1]. The phase boundary between the cartesian and Dedekind columns of Figure 1 is thus exactly the presence of a connection.

4. THE UNIFORM SEPARATION

Theorem 4.1 (Uniform map-level separation). *On each of the De Morgan, Kleene, and Boolean cubical sites, the median collage $\bar{\Phi} : J/K \rightarrow \square^2/K$ of [16] is a test weak equivalence that is not a type weak equivalence. Hence $\mathcal{W}_{\text{type}} \subsetneq \mathcal{W}_{\text{test}}$ on all three sites.*

Theorem 4.2 (Uniform object-level separation). *On each of the three sites of Theorem 4.1, the subdivided mapping cone C of [16] is an explicit finite presheaf that is test-contractible but not type-contractible: the two structures do not agree about which finite objects are aspherical.*

We recall the shape of the De Morgan separation [16]. $K = \{1, \text{sw}, \text{n}, g\}$ is the Klein group acting on the square by the coordinate swap sw , the double reversal $\text{n}(x, y) = (\neg x, \neg y)$, and $g = \text{nsw}$; $W := \square^2/K$. The *join model* J is the double mapping cylinder of the two strata circles, with a twisted K -action, and the *median collage*

$$\Phi = (F(d, a, t), F(d, \neg a, t)), \quad F(d, a, t) := (d \wedge \neg t) \vee (a \wedge t) \vee (d \wedge a),$$

descends to $\bar{\Phi} : J/K \rightarrow W$. The main theorems of [16] state that $\bar{\Phi}$ is a test weak equivalence (both sides realize $B\mathbb{Z}/2 \star B\mathbb{Z}/2$) but not a type weak equivalence (the *nullity theorem*: every map from W into a fibrant replacement of J/K is null-homotopic), and that a subdivided mapping cone C of $\bar{\Phi}$ is test-contractible but not type-contractible. All constructions make sense verbatim on the Kleene and Boolean sites: the formulas for the actions, the join model, and F use only $\wedge, \vee, \neg, 0, 1$.

4.1. The audit.

Lemma 4.3 (Subvariety inheritance). *Every term identity of the De Morgan variety holds in the Kleene and Boolean varieties. In particular the self-duality $\neg F(d, a, t) = F(\neg d, \neg a, t)$, the flow identities $F(d, a, t \wedge s) = F(d, F(d, a, t), s)$ and $F(d, a, t \vee s) = F(F(d, a, t), a, s)$, the equivariance identities of the collage, and the sector formulas of the subdivided cone hold on all three sites.*

Proof. Kleene and Boolean algebras are subvarieties of De Morgan algebras: each is axiomatized by the De Morgan axioms plus further identities/inequalities. Identities valid in a variety hold in every subvariety. $\square$

Lemma 4.4 (No self-negated elements, all three sites). *On each site, no element of a free algebra equals its own negation. Consequently the double reversal n acts freely on the nonempty cells of the square, at every level.*

Proof. De Morgan and Kleene: the all- $(1, 0)$ point p_0 of the literal cube (each pair equal to $(1, 0)$) satisfies $\rho p_0 = p_0$ and is unmixed, hence lies in U_n ; a self-negated φ would satisfy $\varphi(p_0) = 1 - \varphi(p_0)$. Boolean: negation is the output complement, so $\neg\varphi \neq \varphi$ pointwise. $\square$

Lemma 4.5 (First parity lemma, all three sites). *No φ in the free algebra on two generators satisfies $\varphi^{\text{sw}} = \neg\varphi$.*

Proof. De Morgan and Kleene: the point $q = (1, 0, 1, 0)$ satisfies $\text{sw } q = q$ and $\rho q = q$, and is unmixed, hence lies in U_2 ; at q the condition reads $\varphi(q) = 1 - \varphi(q)$. Boolean: at a diagonal input (a, a) the condition reads $\varphi(a, a) = 1 - \varphi(a, a)$. $\square$

Lemma 4.6 (Second parity lemma, all three sites). *No φ in the free algebra on two generators satisfies both $\varphi^{\text{sw}} = \varphi$ and $\varphi^{\text{n}} = \neg\varphi$.*

Proof. De Morgan and Kleene: the point $m = (1, 0, 0, 1)$ is unmixed and its bitwise complement equals $\text{sw } m$; symmetry gives $\varphi(\bar{m}) = \varphi(m)$ while

the second condition gives $\varphi(\overline{m}) = 1 - \varphi(m)$. Boolean: at the input $(0, 1)$, the $\mathbf{n}$ -condition reads $\varphi(1, 0) = 1 - \varphi(0, 1)$ while symmetry gives $\varphi(1, 0) = \varphi(0, 1)$. $\square$

Proof of Theorems 4.1 and 4.2. We audit the proofs of the corresponding theorems of [16], replacing the De Morgan site by the Kleene or Boolean site throughout, and verify that every ingredient remains valid.

Site-uniform machinery. The following results of [16] are proved for an arbitrary presheaf category over a strict test cube category whose interval theory contains $\wedge, \vee, \neg$ and diagonals, and use no further structure of the free algebras: the descent lemma for uniform fillers and the algebraic fibrant replacement with its coequalizer presentation (the proofs use cellular decompositions of monomorphisms, pullback of boxes to representables, and strict naturality — all site-independent); the strictification of maps out of strict quotients (Yoneda plus levelwise colimits); the equivariant filler slide (the slide formulas $\tau \wedge \neg z, \tau \vee z$ exist by Lemma 4.3); the upper bound (U); the equivariant descent dictionary and the sector, pasting, and first-exit lemmas of the object-level proof (coequalizer relations plus diagonals, which all our sites possess); and the collage formula for nerves of sieve-cosieve decompositions (general category theory [4, 20]).

Term identities. Every algebraic identity invoked — self-duality and flow for F , the equivariance of Φ and of the cone slide, the face computations — holds by Lemma 4.3.

The test side. The stratification of the deck action on the cells of the square is site-uniform once Lemma 4.4 is in place: $\mathbf{sw}$ -fixed cells are the diagonal ones (φ, φ) , g -fixed cells the antidiagonal ones $(\varphi, \neg\varphi)$, and $\mathbf{n}$ acts freely; the strata are copies of the interval quotient $\square^1/\neg$, whose category of elements is $B\mathbb{Z}/2$ by the free-covering argument (freeness is again Lemma 4.4); the free parts are contractible covers with the generic cells as terminal objects (stabilizer-freeness of the generic cell: $\mathbf{sw}$ and g move it since $y \neq x$ and $\neg y \neq x$ in the free algebra on two generators, and $\mathbf{n}$ by Lemma 4.4). The collage computation then gives $\mathrm{el}(W) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2$ on each site, and the three-legs argument for $\mathrm{el}(\bar{\Phi})$ being a weak equivalence proceeds verbatim: the collapse-locus closure and flow retraction use only the flow identities and Lemma 4.4; the middle layers use the terminal covers and free quotients. We have verified the stratification and the census ingredients by machine on both new sites (Section 18).

The type side. The nullity theorem's stage induction is site-uniform; its base is the rigidity of invariant cylinder classes of $(J/K)([2])$, which is a formal consequence of Lemmas 4.5 and 4.6 exactly as in [16]: invariance under $\mathbf{sw}$ modulo the deck group forces componentwise $\mathbf{sw}$ -symmetry by Lemma 4.5, and then invariance under $\mathbf{n}$ forces componentwise strict invariance by Lemma 4.6. The machine censuses confirm the rigidity exhaustively: on the Kleene site $(J/K)([2])$ has 96 invariant interior classes, every one with a strictly invariant representative; on the Boolean site, 8, likewise; and on

neither site does any invariant class map to the identity class of W under the collage (no strict sections; on the Boolean site 64 non-invariant triples do hit the identity class — a genuinely Boolean phenomenon — but none is source-invariant). Non-contractibility of W in the type structure follows from (U) and $\text{el}(W) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2$ as in [16].

Assembly. With all ingredients in place, the proofs of the main theorems of [16] go through verbatim on each site: $\bar{\Phi}$ is a test weak equivalence but not a type weak equivalence (Theorem 4.1), and the subdivided mapping cone is test-contractible but not type-contractible (Theorem 4.2). $\square$

Remark 4.7 (What does *not* transfer). The transfer is not a triviality about quotients of proofs: the stage-zero *numerics* differ substantially across the sites (Table 1), and at least one qualitative phenomenon is new on the Boolean site — the existence of (non-invariant) collage preimages of the identity class, which on the De Morgan site is excluded outright. What transfers is the *mechanism*, because its site-specific content was already concentrated in Lemmas 4.5–4.6 by [16].

	De Morgan	Kleene	Boolean
$ F(2) $	168	84	16
classes of $W([2])$	7140	1806	72
strict endomorphism classes	72	38	12
invariant interior classes of $(J/K)([2])$	360	96	8
of which essential	0	0	0
collage-image = id triples	0	0	64
of which source-invariant	0	0	0

TABLE 1. Stage-zero censuses across the three reversal sites (machine-verified; De Morgan values from [16, 18]).

Part 2. The landscape

5. WITNESS MACHINES AND TWISTS

5.1. Witness machines, twists, and the swap character. For a mixed two-stratum subgroup $H \leq B_n$ (two reflections h_1, h_2 with disjoint fixed loci $\square^{m_1}, \square^{m_2}$), the *join model* J_H of [9] has coordinates $m_1 + m_2 + 1$ (the locus parameters and the join parameter t), a twisted H -action ν , and the median collage $\Phi = (F(\ell_1, \ell_2, t)\text{-components})$, $F(d, a, t) = (d \wedge \neg t) \vee (a \wedge t) \vee (d \wedge a)$, descending to $\bar{\Phi}_H : J_H/H \rightarrow \square^n/H$. The *strata-swap character* $\varepsilon : H \rightarrow \mathbb{Z}/2$ takes value 1 on the elements exchanging the two loci; $\nu(h)$ acts on the t -coordinate by $t \mapsto \neg^{\varepsilon(h)} t$. An *invariant interior cell with twist* $\delta \in \text{Hom}(H, H)$ is a cell c of J_H (interior: t -component nonconstant) with $c \circ \sigma_h = \nu(\delta(h)) \cdot c$ for all h ; its class in (J_H/H) is invariant under the source substitutions, i.e., strict-map data for $\square^n/H \rightarrow J_H/H$ by strictification

[16]. An invariant cell is *slide-null* when an invariant homotopy (in one further cylinder variable w) joins its class to a class supported in an end (t -component constant).

6. THE REALIZABILITY DIVIDE

Lemma 6.1 (XOR character law). *In the free Boolean algebra, for $S \subseteq \{1, \dots, n\}$ let $e_S := \bigoplus_{i \in S} x_i$. Then for every signed permutation $h = (\sigma, s)$,*

$$e_S \circ \sigma_h = \neg^{\chi_S(h)} e_{h(S)}, \quad \chi_S(h) := \sum_{i \in S} s_i \bmod 2.$$

In particular, if H stabilizes S then e_S realizes the sign character $\chi_S|_H$.

Proof. $e_S \circ \sigma_h = \bigoplus_{i \in S} (x_{\sigma(i)} \oplus s_i) = e_{\sigma(S)} \oplus \sum_{i \in S} s_i$, and $\oplus 1$ is the output complement. (Verified exhaustively for $n \leq 3$.) $\square$

Theorem 6.2 (Unified realizability, all dimensions). *Let $H \leq B_n$, let $\psi : H \rightarrow \mathbb{Z}/2$ be a character, and let H_ψ denote the ψ -twisted action $h \mapsto \sigma_h \circ \rho^{\psi(h)}$ on the literal cube (ρ = swap-and-complement each pair; on the Boolean site ρ acts trivially on points). A ψ -twisted element ($T \circ \sigma_h = \neg^{\psi(h)} T$) of the free algebra on n generators exists over the site $\mathcal{V}$ if and only if ψ vanishes on the stabilizers, under H_ψ , of every middle point of the site:*

- (1) *Boolean: the vertices of $\{0, 1\}^n$;*
- (2) *Kleene: the proper middle points (every literal pair $(1, 0)$ or $(0, 1)$), which biject with the vertices;*
- (3) *De Morgan: the full middle level of the literal cube, including the mixed points.*

In particular Kleene realizability coincides with Boolean realizability, and the De Morgan site is exceptional exactly because the mixed middle points contribute additional twisted stabilizers.

Proof. Necessity (all sites): evaluating the twisted equation around a cycle of the H_ψ -action at a middle point p gives $T(p) = \psi(\text{cycle}) \oplus T(p)$, so ψ must vanish on $\text{Stab}_{H_\psi}(p)$; middle points are in the domain of the free algebra on each site (for Kleene, the proper points lie in the unmixed poset).

Sufficiency, Boolean: propagate values along H_ψ -orbits of vertices from arbitrary representatives; consistency at a stabilized point is the vanishing hypothesis, and every vertex function is a free Boolean element. When ψ is a stable-set sign character the solution is the signed XOR e_S .

Sufficiency, Kleene (the $\top$ -dominant extension): points of the unmixed poset U_n carry no $(0, 0)$ -pair together with a $(1, 1)$ -pair, so

$$T(p) := \begin{cases} 1 & \text{if } p \text{ has a } (1, 1)\text{-pair,} \\ 0 & \text{if } p \text{ has a } (0, 0)\text{-pair,} \\ t_\psi(p) & \text{if all pairs of } p \text{ are proper,} \end{cases}$$

is well defined, where t_ψ is the Boolean solution read on the proper values. T is monotone: proper points are pairwise incomparable in the literal order

(raising any literal of a proper pair creates a $(1,1)$ -pair), no point above a proper point has a $(0,0)$ -pair, and the outer strata take the extreme values. T is ψ -twisted: substitutions preserve the three strata; ρ exchanges $(1,1)$ - and $(0,0)$ -pairs, so on the outer strata the equation reads $1 = \psi(h) \oplus 0$ exactly when $\psi(h) = 1$; on the proper stratum it is the Boolean equation. (For $n = 2$ the two solutions found by the machine are exactly T and $\neg T$.)

Sufficiency, De Morgan (the weight extension): given a parity-consistent assignment T_0 on the middle level, extend by $T(p) := 0$ for weight $< n$ and $T(p) := 1$ for weight $> n$. Substitutions preserve weight and ρ sends weight w to $2n - w$, so on the outer strata the twisted equation reads $0 = \psi(h) \oplus 1$ precisely when $\psi(h) = 1$, and on the middle level it is the consistency of T_0 . Monotonicity: points of equal weight are incomparable, and the strata are ordered $0 \leq T_0 \leq 1$. The closed form was verified against brute-force enumeration over all subgroups of B_2 and all their characters (25 cases, zero mismatches). $\square$

Remark 6.3. Theorem 6.2 inverts the naive expectation: adding equations to the variety (De Morgan $\rightarrow$ Kleene $\rightarrow$ Boolean) *increases* realizability, because the added equations destroy the carriers of the obstructions. The De Morgan site — the site of cubical type theory — is the exceptional site on which the certificate method works.

	De Morgan	Kleene	Boolean
$ F(2) $	168	84	16
essential invariant classes, $(J_{B_2}/B_2)([2])$	0	18	2
survivors: four-lines / B_3 / H_8 / H_{24}	0/6/5/7	9/19/5/7	9/19/5/7
sign characters realizable	no	yes	yes

TABLE 2. The realizability divide in the machine data (De Morgan values from [9]). “Survivors” are consistent nontrivial twists of the beyond-two-stratum certificate systems; on the Kleene side the counts are exact (monotone-realizability filter over U_3).

7. THE RIGIDITY DISCRIMINANT

Theorem 7.1 (Discriminant). *Let c be an invariant interior cell of J_H with twist δ , and $\psi := \varepsilon \circ \delta$.*

- (1) *If $\psi = 0$, then c is slide-null at stage zero: the slide $t \mapsto t \wedge \neg w$ in the cylinder variable, with the remaining coordinates extended w -constantly, is an invariant homotopy to the $t=0$ end.*
- (2) *If $\psi \neq 0$, then c is rigid at stage zero: no invariant homotopy joins its class to an end class.*

Proof. (1) The slide formulas are term operations available on all three sites; when $\psi = 0$ the deck action of the twist never flips t , so the slide is δ -equivariant, its $w=0$ face is c and its $w=1$ face has constant t -component. (2) Let $h \mapsto k_h$ assign to each generator a deck element witnessing the invariance of a homotopy c_4 with end faces. At the $w=1$ face the t -component is a constant ε_0 ; the twisted equation there reads $\varepsilon_0 = \psi'(h) \oplus \varepsilon_0$ for the homotopy's own character ψ' , forcing $\psi' = 0$. Hence the $w=0$ face has an *untwisted* t -component; but that face is a deck translate of c , whose t -component is ψ -twisted (negation preserves twistedness). For h with $\psi(h) = 1$ and any point p , twistedness gives $T(\sigma_h p) = \neg T(-)$ while untwistedness gives equality — a pointwise contradiction, so no such function, hence no such homotopy, exists. $\square$

Theorem 7.2 (Permanence). *Suppose $\varepsilon \neq 0$ and let c be as in Theorem 7.1(2), with $g_c : \square^n/H \rightarrow J_H/H$ the corresponding strict map. Then the class of g_c in $[\square^n/H, R(J_H/H)]$ differs from the class of every end map, at every stage of the fibrant replacement.*

Proof. The t -coordinate is a strict map $J_H \rightarrow \square^1$, equivariant over the ε -action of H on $\square^1$ by reversal; since $\varepsilon \neq 0$ it descends to

$$q : J_H/H \longrightarrow \mathbb{L} := \square^1/\neg,$$

and both ends of J_H land on the unique vertex of $\mathbb{L}$, so $q(\text{end maps})$ are constant. Suppose $[g_c] = [\text{end}]$; composing with $R(q)$ gives $[q \circ g_c] = [\text{const}]$ in $[\square^n/H, R(\mathbb{L})]$. A type cylinder homotopy is in particular a test homotopy (the unit of the replacement is a type-trivial cofibration, hence a test equivalence), and $\mathbb{L}$ is a $K(\mathbb{Z}/2, 1)$ in the test structure on each site [16], so

$$[\square^n/H, \mathbb{L}]_{\text{test}} \cong \text{Hom}(\pi_1 \text{el}(\square^n/H), \mathbb{Z}/2).$$

The map $q \circ g_c$ classifies the double cover with monodromy ψ : the twisted equation $T(\sigma_h p) = \psi(h) \oplus T(\dots)$ is precisely the statement that transporting around the loop of h in the free part $BH \subseteq \text{el}(\square^n/H)$ interchanges the sheets iff $\psi(h) = 1$. Since $\psi \neq 0$ the class is nontrivial, contradicting $[qg_c] = 0$. $\square$

Remark 7.3. Theorem 7.2 is a type-side obstruction detected by test-side covering theory. On the Boolean Klein machine the projection q factors strictly through the collage: the identity $F(d, a, t) \oplus F(d, \neg a, t) = t$ (verified on all of $\text{FB}(2)^3$) gives $q = t^* \circ \bar{\Phi}$ with $t^* = [x \oplus y]$ the unique nontrivial strict map $\square^2/B_2 \rightarrow \mathbb{L}$.

8. COVERING DESCENT

Throughout this section G is a finite group acting on a presheaf X *freely on nonempty cells*: no nontrivial element fixes a nonempty cell at any level. Write $Y := X/G$ (levelwise orbits); $p : X \rightarrow Y$ is then a levelwise principal G -covering.

Lemma 8.1 (Coverings are local systems). *A presheaf map $E \rightarrow Z$ that is a levelwise principal G -covering is the same data as a functor $\text{el}(Z) \rightarrow G\text{-Tors}$, i.e., a $\pi_1(\text{el}Z)$ -set with free transitive G -action. In particular isomorphism classes of G -coverings of Z are classified by $\text{Hom}(\pi_1 \text{el}(Z), G)$ up to conjugacy, and cylinder-homotopic maps pull coverings back to isomorphic coverings.*

Proof. A presheaf over Z is a presheaf on $\text{el}(Z)$; the levelwise torsor condition says each value is a G -torsor and each structure map a torsor map, which is a local system valued in G -torsors. The classification and homotopy invariance are the standard statements for local systems on a small category, via the nerve. $\square$

Lemma 8.2 (Simply connected box domains). *For every box inclusion $j = (S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$, the category of elements $\text{el}(\text{dom}(j))$ is simply connected.*

Proof. The category-of-elements construction preserves colimits of presheaves, so the pushout $\text{dom}(j) = (S \times \square^1) \cup_{S \times \{\varepsilon\}} (\square^n \times \{\varepsilon\})$ gives a pushout of categories along monic inclusions, hence a homotopy pushout of nerves. The leg $\text{el}(S \times \{\varepsilon\}) \hookrightarrow \text{el}(S \times \square^1)$ is a π_1 -isomorphism (the cylinder projection is a test equivalence and the end inclusion is its section), and $\text{el}(\square^n)$ is contractible (terminal generic cell). Van Kampen gives $\pi_1 \cong \pi_1(\text{el}S) *_{\pi_1(\text{el}S)} 1 = 1$, and the pushout is connected. $\square$

Lemma 8.3 (Covering extension). *Let $Z \hookrightarrow Z'$ be a relative cell complex whose attachments are boxes of the uniform structure (e.g. the algebraic fibrant replacement $Z' = R(Z)$ of [16]). Then the restriction functor*

$$\text{Cov}_G(Z') \longrightarrow \text{Cov}_G(Z)$$

is an equivalence of groupoids. In particular every G -covering of Z extends to Z' , uniquely up to unique isomorphism relative to Z ; and the total object P of the extension of a free quotient $p : X \rightarrow Y$ along $Y \hookrightarrow R(Y)$ is a uniformly fibrant object containing X , with free G -action, $P/G \cong R(Y)$, and $X \hookrightarrow P$ a type-trivial cofibration — a G -equivariant fibrant replacement of X realizing the replacement of the quotient.

Proof. G -coverings of a presheaf are local systems valued in G -torsors on its category of elements (Lemma 8.1). For a single box attachment $Z' = Z \cup_b \text{cod}(j)$, the category of elements is the pushout $\text{el}(Z') = \text{el}(Z) \cup_{\text{el}(\text{dom } j)} \text{el}(\text{cod } j)$, so

$$\text{Cov}_G(Z') \simeq \text{Cov}_G(Z) \times_{\text{Cov}_G(\text{dom } j)} \text{Cov}_G(\text{cod } j).$$

Now $\text{cod}(j) = \square^n \times \square^1$ is representable, so $\text{el}(\text{cod } j)$ is contractible and its covering groupoid is that of a point; and $\text{el}(\text{dom } j)$ is simply connected (Lemma 8.2), so every local system on it is trivializable, and its trivializations form a contractible groupoid. Hence the fiber-product factor is a

contractible correction and the projection $\text{Cov}_G(Z') \rightarrow \text{Cov}_G(Z)$ is an equivalence. Transfinite compositions and retracts of attachments pass to limits of groupoid equivalences.

For the second statement, let $P \rightarrow R(Y)$ be the essentially unique extension of $X \rightarrow Y$. Levelwise $P/G \cong R(Y)$ and the G -action is free away from nothing (torsor fibers). $X \hookrightarrow P$ is a transfinite composition of pushouts of boxes: over each downstairs attachment the covering restricts on $\text{dom}(j)$ to a trivialized system by Lemma 8.2, so its total object is G disjoint copies of $\text{dom}(j)$, and the extension attaches G disjoint lifted boxes. Fibrancy of P : fill a box of P by projecting to $R(Y)$, filling uniformly, and lifting the filler along the covering matching the given lift at the ε -end generator; the lift agrees with the given data on all of $\text{dom}(j)$ because two lifts of a map from a simply connected domain agreeing at one cell agree everywhere, and strict naturality is preserved since restriction preserves the ε -end base cell. $\square$

Theorem 8.4 (Covering descent). *Let G act freely (on nonempty cells) on X and X' , let $f : X \rightarrow X'$ be equivariant, and suppose the quotient map $\bar{f} : X/G \rightarrow X'/G$ is a type weak equivalence. Then f is a type weak equivalence. Contrapositively, non-equivalence descends: if $f \notin \mathcal{W}_{\text{type}}$ then $\bar{f} \notin \mathcal{W}_{\text{type}}$.*

Proof. Write $Y = X/G$, $Y' = X'/G$, and let P, P' be the equivariant replacements of Lemma 8.3. The functorial replacement $R(\bar{f}) : R(Y) \rightarrow R(Y')$ pulls the covering P' back to a covering of $R(Y)$ which restricts on Y to $\bar{f}^*(X' \rightarrow Y') \cong (X \rightarrow Y)$ (equivariance of f); by the uniqueness of Lemma 8.3 this pullback is P , giving a canonical equivariant $F : P \rightarrow P'$ over $R(\bar{f})$ with $F|_X = f$ up to the identification. Since $\bar{f} \in \mathcal{W}_{\text{type}}$ and all objects are cofibrant with $R(Y), R(Y')$ fibrant, $R(\bar{f})$ has a homotopy inverse $\bar{g}$ with cylinder homotopies $\bar{H} : \bar{g}R(\bar{f}) \simeq \text{id}$ and $\bar{K} : R(\bar{f})\bar{g} \simeq \text{id}$. By Lemma 8.1 the coverings are classified by characters of $\pi_1 \text{el}$, homotopy-invariantly; from $R(\bar{f})\bar{g} \simeq \text{id}$ we get $\bar{g}^*[P] = \bar{g}^*R(\bar{f})^*[P'] = [P']$, so $\bar{g}$ lifts to an equivariant $\tilde{g} : P' \rightarrow P$. Similarly the homotopies lift (a cylinder $Z \times \square^1$ is connected when Z is, so extensions of coverings along cylinders are again unique), each lift agreeing with the expected end maps up to a deck translation. Thus $\tilde{g}F \simeq \gamma_a$ and $F\tilde{g} \simeq \gamma_b$ for deck automorphisms γ_a, γ_b ; as these are isomorphisms, F has a left and a right homotopy inverse, hence is a homotopy equivalence. Since $X \hookrightarrow P$ and $X' \hookrightarrow P'$ are type-trivial cofibrations and F extends f , two-out-of-three gives $f \in \mathcal{W}_{\text{type}}$. $\square$

Remark 8.5. The test-side counterpart is classical: el of a free quotient is the quotient of el , and free quotients of equivariant test equivalences are test equivalences (the Borel argument of [16]). Theorem 8.4 is its type-side partner, and the two together transfer an attached separation ($\bar{\Phi}_N \in \mathcal{W}_{\text{test}} \setminus \mathcal{W}_{\text{type}}$) from N to H along any tower.

9. SITE-INVARIANCE OF THE TEST SIDE

Lemma 9.1 (el-invariance). *For any two reversal sites and any $H \leq B_n$, the categories of elements $\text{el}(\square^n/H)$ (and those of the witness machines) are connected by weak equivalences determined by the common stratification data: the isotropy poset of H , the fixed-locus patterns, and the residual actions — all of which depend only on the signed-cycle combinatorics of H .*

Proof. We argue by induction on n ; all ingredients are those of the self-containedness appendix of [16], checked to depend only on the group data.

The stratification is site-independent. A cell of $\square^n$ is stabilized by $h \in H$ iff its components satisfy the fixed-cell equations of h ; by the locus-pattern analysis (valid on each site because distinct signed generators are distinct in every free algebra, and no element is self-negated), the fixed locus of a subgroup $S \leq H$ is either empty or a subcube $\square^{d(S)}$ embedded by the pattern $x_i \mapsto \neg^{\varepsilon_i} m_{\text{cl}(i)}$, where the classes, signs, and dimension $d(S)$ are determined by the signed-cycle combinatorics of S alone. Hence the poset of isotropy types, the loci, their intersections (loci of joins), and the residual actions of normalizers are the same on all sites.

The collage is site-independent. Decompose $\text{el}(\square^n/H)$ by the sieve of stabilized cells and its free complement, as in [16]: (i) the free part has a contractible cover — the full subcategory of stabilizer-free cells of $\square^n$, with the generic cell terminal — on which H acts freely, so the piece is BH ; (ii) the sieve is the union of the strata, each stratum the quotient of a locus $\square^{d(S)}$ by its residual group — an isotropy quotient in dimension $d(S) < n$, equivalent across the sites by the inductive hypothesis (base $n = 0$: points); overlaps of strata are again loci, handled by the same induction over the isotropy poset; (iii) each comma piece is an arrow category fibered over its stratum with contractible fibers (coslices under the terminal generic cell), contractible free cover, and quotient $B(\text{Stab})$ by Quillen's Theorem A — all identified by the group data. The collage formula of [4, 20] assembles the pieces, and every piece and gluing map has been matched across the sites, so the nerves are connected by weak equivalences as claimed. The same induction applies verbatim to the witness machines, whose cells are built from cube cells, loci, and the combinatorial E_N -factor. $\square$

10. THE COLLAGE FORMULA

The decompositions used above assemble into a single closed formula, which both organizes the earlier special cases and makes the el-invariants of isotropy quotients exactly computable.

Theorem 10.1 (Collage formula). *Let $H \leq B_n$ and let $\Sigma \subseteq \square^n$ be the sieve of cells with nontrivial stabilizer. Then*

$$\text{Nel}(\square^n/H) \simeq \text{hocolim} \left(\text{el}(\Sigma/H) \longleftarrow (\text{el } \Sigma)_{hH} \longrightarrow BH \right),$$

where $(-)_hH$ is the Borel construction. Moreover $\mathrm{el}(\Sigma)$ is equivalent to the nerve of the incidence poset of the isotropy loci, so the middle term is the classifying space of a finite complex of groups whose vertex groups are the setwise stabilizers of the loci.

Proof. Decompose $\mathrm{el}(\square^n/H)$ along the closed sieve Σ/H and its open complement (the free cells): the nerve is the homotopy pushout of the sieve part, the cosieve part, and the comma category of arrows from the sieve into the cosieve. The sieve part is $\mathrm{el}(\Sigma/H)$. The cosieve part has the full subcategory of stabilizer-free cells of $\square^n$ as a cover, which is contractible (the generic cell is terminal) and carries a free H -action, so the piece is BH . The comma is $\mathrm{Arr}(\Sigma, F)/H$ with F the free part upstairs; projecting to the source, each fiber is a coslice with terminal object (the generic cell), so by Quillen's Theorem A the comma is equivalent to $\mathrm{el}(\Sigma)//H = (\mathrm{el} \Sigma)_{hH}$, the action on $\mathrm{el}(\Sigma)$ being honest and the F -component making it free. The legs are the collapse $(\mathrm{el} \Sigma)_{hH} \rightarrow BH$ and the natural projection $(\mathrm{el} \Sigma)_{hH} \rightarrow \mathrm{el}(\Sigma/H)$. For the addendum: each locus and each intersection of loci (the locus of the join of the stabilizers) is a subcube, hence has contractible el ; by the nerve-of-cover lemma, $\mathrm{el}(\Sigma)$ is equivalent to the nerve of the incidence poset of the loci, and the Borel construction of a poset of contractible pieces with setwise stabilizers is the complex of groups stated. $\square$

$$\begin{array}{ccc}
 (\mathrm{el} \Sigma)_{hH} & \xrightarrow{\text{collapse}} & BH \\
 \text{proj} \downarrow & & \downarrow \\
 \mathrm{el}(\Sigma/H) & \xrightarrow{\quad \quad \quad} & N\mathrm{el}(\square^n/H)
 \end{array}
 \quad \begin{array}{c} \\ \uparrow(\mathrm{ho}) \end{array}$$

FIGURE 2. The collage formula (Theorem 10.1): the nerve of an isotropy quotient is the homotopy pushout of the sieve quotient, the Borel construction of the reflection arrangement, and BH . When $\mathrm{el} \Sigma \simeq *$ the right leg is an equivalence and the sieve inclusion induces $\mathrm{el}(\Sigma/H) \simeq \mathrm{el}(\square^n/H)$ (Proposition 14.9).

Remark 10.2. The formula (Figure 2) reduces every $\mathrm{el}(\square^n/H)$ — and, applied to the witness machines, every el appearing in this paper — to finite group cohomology plus finite poset data. All the $\mathbb{F}_2$ -cochain computations in Proposition 15.11 below are exact evaluations of this formula (normalized bar complexes for the group pieces, iterated homotopy pushout cochain models for the gluings; artifact).

Part 3. The criterion

11. REFLECTIONS, VERTICES, AND AUTOMATIC HYPOTHESES

Two elementary facts about reflections anchor everything.

Lemma 11.1. *For a nontrivial element of B_n the following are equivalent: (a) it fixes a nonempty cell; (b) every cycle of its signed cycle decomposition has even sign sum; (c) it fixes a vertex. In particular the two descriptions of reflection in Definition 2.1 agree.*

Proof. (b) $\Rightarrow$ (c): on a cycle of even sign sum, propagate a vertex value around the cycle: the total twist is the sign sum, which is even, so a compatible assignment exists on each cycle. (c) $\Rightarrow$ (a): a vertex is a nonempty cell. (a) $\Rightarrow$ (b): a fixed cell forces, on each cycle, a fixed element of the free algebra whose propagation around the cycle twists by the sign sum; an odd twist would force the element to equal its own negation, impossible in every free algebra of the reversal varieties. $\square$

Lemma 11.2 (Odd order). *Every element of odd order in B_n is a reflection or the identity. Consequently, for every $H \leq B_n$ the quotient $H/\langle\langle \text{Refl}(H) \rangle\rangle$ is a 2-group; in particular, if it is nontrivial it has a $\mathbb{Z}/2$ -quotient.*

Proof. A signed cycle of length ℓ and sign s has order ℓ if $s = 0$ and 2ℓ if $s = 1$; an element of odd order therefore has all cycle signs zero, i.e., is a reflection or the identity. Hence all odd-order elements of H lie in $\langle\langle \text{Refl}(H) \rangle\rangle$, so the quotient has no elements of odd prime order, and a nontrivial finite 2-group surjects onto $\mathbb{Z}/2$. $\square$

Lemma 11.3 (Automatic character triviality). *Let ψ be a character of H realizable as a twisted element of the free algebra on any of the three sites ($T \circ \sigma_h = \neg^{\psi(h)} T$). Then ψ vanishes on every reflection, hence on $\langle\langle \text{Refl}(H) \rangle\rangle$. In particular reflection-generated groups have no nontrivial realizable characters, on any reversal site.*

Proof. Evaluating the twisted equation at a vertex v fixed by h gives $T(v) = \psi(h) \oplus T(v)$ pointwise (Boolean; on the De Morgan and Kleene sites evaluate at the corresponding proper literal point, which is ρ -compatible), so $\psi(h) = 0$ for h in a vertex stabilizer. Apply Lemma 11.1(i). $\square$

12. CERTIFICATES ARE EXACTLY REFLECTION GENERATION

We work with the exact certificate system of Part 2: for a mixed H with a reflection-free normal $V \trianglelefteq H$ and $G = H/V$, and a twist $\delta \in \text{Hom}(H, H)$, the *u-part* asks for an equivariant cell $c : \square^n \rightarrow \square^n$ with $c \circ \sigma_h = \delta(h) \circ c$, and the *e-part* for a G -valued vertex function E with $E(h \cdot v) = \bar{\delta}(h) \cdot E(v)$; on the Boolean site both systems are exact (parity-consistent assignments are cells), and on the Kleene site exactness holds after the monotone-realizability filter. H is *certified* when no nontrivial δ is consistent.

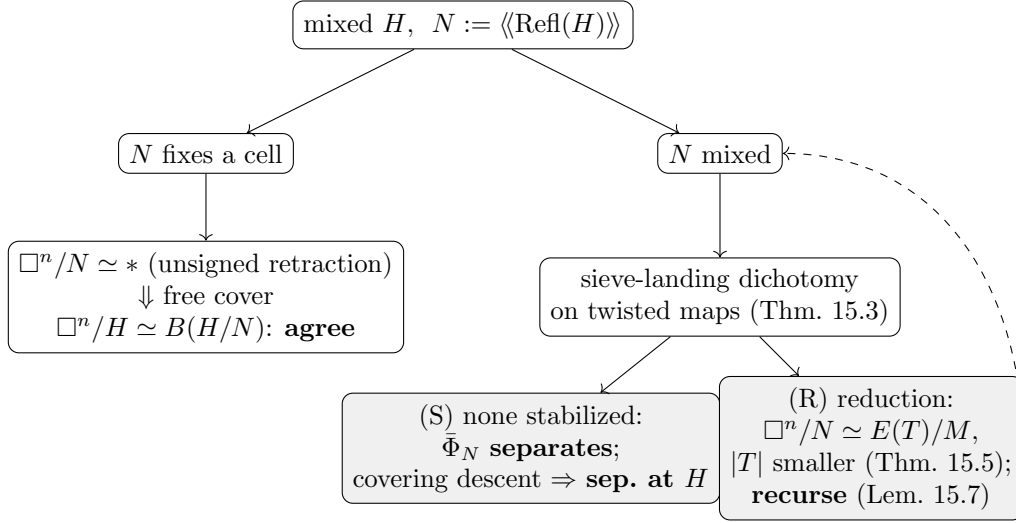


FIGURE 3. The criterion at every n (Theorem 15.8): the two halves, the dichotomy, and the well-founded recursion (dashed).

Theorem 12.1. H is certified if and only if $H = \langle\langle \text{Refl}(H) \rangle\rangle$.

Proof. ($\Leftarrow$) Let δ be consistent. The u-part cell restricts to a vertex map $\hat{c}$ with $\hat{c}(h \cdot v) = \delta(h) \cdot \hat{c}(v)$, so $\delta(\text{Stab}_H(v)) \subseteq \text{Stab}_{B_n}(\hat{c}(v))$. The e-part, since G acts freely on itself, forces $\bar{\delta}(\text{Stab}_H(v)) = 1$, i.e. $\delta(\text{Stab}_H(v)) \subseteq V$. But V is reflection-free while nontrivial vertex stabilizers consist of reflections (Lemma 11.1(ii)), so $\delta(\text{Stab}_H(v)) = 1$ for every vertex v . By Lemma 11.1(i) every reflection lies in some vertex stabilizer, so δ kills $\text{Refl}(H)$, hence the normal closure ($\ker \delta$ is normal), hence H ; thus δ is trivial.

($\Rightarrow$) Suppose $N := \langle\langle \text{Refl}(H) \rangle\rangle \subsetneq H$. By Lemma 11.2 the quotient H/N is a nontrivial 2-group, so there is a character $\psi : H \rightarrow \mathbb{Z}/2$ vanishing on N and not on H . By Cauchy's theorem H contains an involution k ; the vertex action of B_n is faithful, so k moves some vertex v_0 . Then $\delta(h) := k^{\psi(h)}$ is a well-defined homomorphism $H \rightarrow H$ (because $k^2 = 1$), nontrivial since $\psi \neq 0$. Since ψ vanishes on all vertex stabilizers (they consist of reflections, which lie in N), there is a ψ -twisted vertex-level function $T : \{0, 1\}^n \rightarrow \mathbb{Z}/2$ with $T(h \cdot p) = \psi(h) \oplus T(p)$ (propagate along orbits; consistency at stabilized points is exactly the vanishing). The *multiplexer cell* $c(p) := k^{T(p)} \cdot v_0$ — encoded coordinatewise as a Boolean cell — satisfies $c(\sigma_h p) = \delta(h) \cdot c(p)$, and $E(h \cdot v_0) := \bar{\delta}(h)$ is well-defined on the orbit of v_0 (as $\bar{\delta}$ kills the stabilizer) and extends freely elsewhere. Hence δ is a consistent nontrivial twist and H is not certified. $\square$

Remark 12.2. Theorem 12.1 explains the empirical tables of Part 2 at a glance (14 groups across $n = 2, 3, 4$, all matching), identifies the De Morgan certificates of [9] as instances (there the realizability filter is strictly stronger,

but reflection-generation still implies certification by the same argument, and Lemma 11.3 discharges the character hypothesis), and turns “certified” from a computation into a group-theoretic property.

13. THE AGREEMENT HALF

Theorem 13.1 (Two-step covering agreement). *Let H be mixed and suppose $N = \langle\langle \text{Refl}(H) \rangle\rangle$ has a common fixed cell. Then, on each reversal site, $\square^n/N$ is contractible in both structures, H/N acts freely on its cells, and*

$$\square^n/H \simeq B(H/N)$$

in both structures. In particular $\square^n/H$ agrees.

Proof. Step 1. The common-fixed-cell condition for N is a consistent parity system on the coordinates, whose classes and signs define the fixed locus $L \cong \square^d$ with $d < n$ (a nontrivial reflection has a cycle of length ≥ 2 , so some class has size ≥ 2). N acts *trivially* on L : an element of N fixes every cell of L by definition of the common fixed pattern. Consequently the induced action of N on the classes carries no signs, and the consensus retraction

$$(R_t)_i = F(x_i, \neg^{\varepsilon_i} M_{\text{cl}(i)}, t), \quad M_k := \bigwedge_{j \in \text{class } k} \neg^{\varepsilon_j} x_j,$$

with F the median interpolation of [16], is N -equivariant — the only equivariance conditions are permutations *within* and *among* equal-size classes without signs, which the symmetric M satisfies — and no self-duality is required anywhere. R descends to the quotient and contracts $\square^n/N$ onto $L/N = L \cong \square^d$, strictly; hence $\square^n/N$ is contractible in both structures.

Step 2. H normalizes N , so acts on $\square^n/N$ through H/N . The action is free on nonempty cells: if $h \cdot c = \nu \cdot c$ for some $\nu \in N$ then $\nu^{-1}h$ fixes the cell c , hence is a reflection or the identity, hence lies in N , so $h \in N$. By the covering theory of Part 2 (equivariant replacement of the free quotient; on the test side, el of the free quotient is the Borel construction), the quotient of a both-structures contractible object by a free action of the finite group H/N is a $B(H/N)$ in both structures — the argument of the free family of [9], applied verbatim with the contractible base $\square^n/N$ in place of $\square^n$. Finally $\square^n/H = (\square^n/N)/(H/N)$ since quotients are levelwise. $\square$

Remark 13.2. Theorem 13.1 subsumes the median-retraction family of [9]: for $H \leq \Sigma_n \times \langle z \rangle$ mixed, the reflections generate a subgroup fixing the diagonal-type locus, and the two-step argument applies with no parity condition on n . It also settles the five “edge” groups of dimension 4 found by the machine search — degenerate groups admitting *no* equivariant consensus pattern with odd negated classes — for which any one-step equivariant retraction provably fails: the two-step argument sidesteps the sign obstruction entirely, because all signs live in H/N and are absorbed by the covering step.

14. THE SEPARATION HALF: THE UNIFORM CYLINDER MACHINE

Let N be mixed and reflection-generated, on any reversal site. Write $\Sigma \subseteq \square^n$ for the *stabilized sieve*: the subpresheaf of cells with nontrivial stabilizer — the union of the fixed loci of the reflections of N , including all their intersections and nestings — and E_N for the 0-coskeletal codiscrete nerve of N ($E_N([m]) = N^{\text{vertices}([m])}$), contractible with N acting freely by translation.

Construction 14.1 (The cylinder machine).

$$J_N := (\square^n \times E_N) \cup_{\Sigma \times E_N \times \{0\}} (\Sigma \times E_N \times \square^1) / \sim, \quad (x, e, 1) \sim (x, e', 1),$$

the mapping cylinder of the projection $\Sigma \times E_N \rightarrow \Sigma$ glued into the body along its source. N acts diagonally; the action is free away from the collapsed end (on the body and the open cylinder because it is free on E_N ; the collapsed end is the bare Σ with its usual action). The projection $\Phi : J_N \rightarrow \square^n$ (forget E_N and the cylinder coordinate) is equivariant and descends to $\bar{\Phi}_N : J_N/N \rightarrow \square^n/N$.

$$\begin{array}{ccccc} \Sigma \times E_N \times \{0\} & \hookrightarrow & \square^n \times E_N & & \\ \downarrow & & \downarrow & & \\ (\Sigma \times E_N \times \square^1) / \sim & \longrightarrow & J_N & \xrightarrow{\Phi} & \square^n \end{array}$$

FIGURE 4. The uniform cylinder machine (Construction 14.1): the body $\square^n \times E_N$ (free, contractible), the cylinder over the stabilized sieve with collapsed end, and the equivariant projection Φ inducing $\bar{\Phi}_N : J_N/N \rightarrow \square^n/N$.

A single cylinder over the whole stabilized sieve (Figure 4) replaces the per-stratum horns of [9]: orbits, intersections, and nestings of loci require no case distinctions.

Proposition 14.2 (Test side). *$\text{el}(\bar{\Phi}_N)$ is a weak equivalence, for every mixed N .*

Proof. Decompose both sides by the stabilized/free dichotomy. On the target, the sieve is $\text{el}(\Sigma/N)$ and the cosieve is the free part, whose cover is contractible with terminal generic cell, so the cosieve piece is BN . On the source, the sieve consists of the cylinder cells; the cylinder direction is a strict deformation retraction onto the collapsed end, which is the bare Σ , so the sieve piece is again $\text{el}(\Sigma/N)$ and $\bar{\Phi}_N$ restricts to the *identity* on it. The source cosieve is (free cells) $\times E_N$ (before the collapse), whose cover is contractible (product of a category with terminal object and the contractible E_N) with free N -action, so the piece is BN , and Φ induces the identity of N on fundamental groups. The comma pieces compare by the

same contractible-cover argument (Quillen's Theorem A over the sieve, as in Part 2), and the collage formula finishes. $\square$

Theorem 14.3 (Type side: the trichotomy). *Suppose N is reflection-generated. Then every element of $[\square^n/N, R(J_N/N)]$ is homotopic to one of: (a) a constant; (b) a class factoring through the stabilized sieve $\Sigma/N \subseteq J_N/N$ (the collapsed end); or (c) an invariant-polynomial class — a body class $(c, \bar{e})$ whose cube component c is strictly invariant ($c \circ \sigma_h = c$ for all $h \in N$; equivalently, the components of c are N -invariant elements of the free algebra) and whose E_N -component is orbitwise constant.*

Proof. By the strictification and equivariant-descent dictionary of [16], classes in $[\square^n/N, R(J_N/N)]$ are invariant cells of the algebraic replacement at level $[n]$, modulo chains of invariant homotopies one level up, and the replacement splits into *sector subreplacements* along the two closed sieves of J_N :

$$\bar{B} := \square^n \times E_N, \quad \bar{C} := (\Sigma \times E_N \times \square^1)/\sim, \quad \bar{B} \cap \bar{C} = \Sigma \times E_N \times \{0\},$$

the *body*, the *cylinder*, and their *interface*. Every cell of J_N is supported in $\bar{B}$ or in $\bar{C}$ (a cylinder cell has its τ -faces at the interface and at the collapsed end; no cell meets both the body beyond the interface and the end), and the sector bookkeeping for boxes, fillers and reindexings is exactly that of the sector lemmas of [16], whose proofs use only the coequalizer presentation of the replacement and the presence of diagonals.

The cylinder sector. The slide $(x, e, \tau) \mapsto (x, e, \tau \vee z)$ is a totally defined, N -equivariant strict homotopy of $\bar{C}$ onto the collapsed end, compatible with fillers by the uniformity relations (the equivariant filler slide of [16]). Hence every invariant class supported in the cylinder subreplacement — at every stage — is homotopic to a class supported in the collapsed end, i.e., factors through Σ/N : type (b).

The body sector. Here no slide is needed: the body is handled by covering theory. N acts freely on $\bar{B} = \square^n \times E_N$ (translation on E_N), and $\bar{B}$ is contractible in both structures (E_N is 0-coskeletal, $\square^n$ representable). By Lemma 8.3, the replacement of $\bar{B}/N$ is the free quotient P/N of an equivariant replacement $P \simeq *$, and by Lemma 8.1 together with the equivariant Whitehead argument for free actions, homotopy classes of maps $\square^n/N \rightarrow R(\bar{B}/N)$ are classified by their pullback coverings:

$$[\square^n/N, R(\bar{B}/N)] \cong \text{Cov}_N(\square^n/N)_{/\cong} \cong \text{Hom}(\pi_1 \text{el}(\square^n/N), N)/\text{conj}.$$

In particular every body-sector class, at every stage, is homotopic to a stage-zero body class. Stage-zero body classes are pairs (cube component with twist δ , E_N -component), and their consistency is precisely the exact certificate system: the cube component is the u-part and the E_N -component the e-part. By Theorem 12.1 only $\delta = 1$ occurs ($\delta = \text{id}$ -like twists are killed by the e-part, since vertex stabilizers are nontrivial and act freely on E_N); and $\delta = 1$ forces the cube component to be strictly invariant and the E_N -component to be orbitwise constant — type (c) — or the class is degenerate

onto a point — type (a). Distinct orbitwise-constant E_N -labelings are joined by invariant E_N -edges (the codiscrete 1-cells, chosen orbitwise), so the class of type (c) depends only on the strictly invariant cube component c .

Mixed boxes and chains. A fresh filler of a box supported in one sector lies in that sector's subreplacement. A mixed box pastes through the interface by the sector-pasting lemma of [16] (its proof uses only the coequalizer relations and connections, both available): the chain through a mixed filler is replaced by a restricted step to an interface class followed by cylinder material. Interface classes are cylinder classes, hence of type (b). The first-exit induction of [16], with the sector pair $(\bar{B}, \bar{C})$, therefore reduces every invariant chain to one ending in a class of type (a), (b), or (c). $\square$

Hypothesis 14.4 (Non-domination). *For every reflection-generated mixed N : (i) $\text{el}(\square^n/N)$ is not a homotopy retract of $\text{el}(\Sigma/N)$; and (ii) the identity of $\square^n/N$ is not homotopic, in the type structure, to any strictly invariant endomorphism class (the images of the invariant-polynomial classes under $\bar{\Phi}_N$).*

Remark 14.5. Part (ii) has a test-side necessary reduction: a homotopy $\text{id} \simeq \bar{\Phi}_N \circ f$ with f a body class factors the identity through $\bar{B}/N$, which is a BN -object by the covering classification above, so (ii) fails only if $\text{el}(\square^n/N)$ is a homotopy retract of BN — a third non-domination condition, decidable by el -data. Where rank comparisons do not settle it, (ii) is a stage-zero-visible type-side condition: on the Boolean site the strict endomorphism classes and the strict-homotopy component of the identity are finite, exactly computable sets, and for the Klein machine the component is $\{\text{id}\}$ while the identity is not strictly invariant; the artifact verifies (ii) at stage zero for the $n \leq 3$ bases.

Theorem 14.6 (Fundamental group of an isotropy quotient). *For every $H \leq B_n$, the group $\pi_1 \text{el}(\square^n/H)$ is a quotient of H in which the stabilizer of every cell dies. In particular, if H is generated by elements with nonempty fixed locus — e.g. if H is reflection-generated — then $\text{el}(\square^n/H)$ is simply connected.*

Proof. Cells of $\square^n/H$ are H -orbits of cells, and a morphism $[z] \rightarrow [x]$ of $\text{el}(\square^n/H)$ is a cube map u with $[x] \cdot u = [z]$. *Edges lift:* given a representative z' of $[z]$ and such a u , we have $x \cdot u = \sigma_h z'$ for some representative x and h , so $\sigma_{h^{-1}} x \cdot u = z'$; iterating along an edge path, every edge loop of $\text{el}(\square^n/H)$ at a base cell b lifts to an edge path in $\text{el}(\square^n)$ from a representative b' to $\sigma_h b'$ for a unique $h \in H$. Since $\text{el}(\square^n)$ is contractible (the generic cell is terminal), two lifts with the same h are homotopic, the concatenation of loops multiplies the h 's, and every h occurs (project any path from b' to $\sigma_h b'$). Hence $h \mapsto [\text{projected loop}]$ is a surjection $H \twoheadrightarrow \pi_1 \text{el}(\square^n/H)$. If h stabilizes some cell x , conjugate the basepoint to $[x]$: the lift of the corresponding loop can be taken constant at x (since $\sigma_h x = x$), so the loop is constant and h dies; a general element of the normal closure of the stabilizers dies because the

kernel is normal. When H is generated by stabilizing elements the surjection kills a generating set, and reflections stabilize vertices (Lemma 11.1), giving the last claim. $\square$

Proposition 14.7 (Cover lifting). *Let N be reflection-generated and suppose the identity of $\square^n/N$ factors through Σ/N up to homotopy. Then $\text{el}(\square^n/N)$ is a homotopy retract of the universal cover of one component of $\text{el}(\Sigma/N)$. Consequently, if every component of $\text{el}(\Sigma/N)$ has an $\mathbb{F}_2$ -acyclic universal cover, then Hypothesis 14.4(i) follows from $\tilde{H}^*(\text{el}(\square^n/N); \mathbb{F}_2) \neq 0$ alone.*

Proof. A factorization $\text{id} \simeq q \circ g$ with $q: \Sigma/N \hookrightarrow \square^n/N$ induces $s := \text{el}(g)$ with $\text{el}(q) \circ s \simeq \text{id}$. The source is simply connected by Theorem 14.6, so s lifts along the universal covering $\pi: \tilde{E} \rightarrow E$ of the component E it hits: $\pi \circ \tilde{s} = s$. Then $(\text{el}(q) \circ \pi) \circ \tilde{s} \simeq \text{id}$, the claimed retraction; a retract of an $\mathbb{F}_2$ -acyclic space is $\mathbb{F}_2$ -acyclic, and the constant case is the empty-sieve instance of the same argument. $\square$

Remark 14.8. Theorem 14.6 is machine-confirmed: in the vertex Čech model, $H_1(\text{el}(\square^n/N); \mathbb{F}_2) = 0$ for every reflection-generated mixed class with $n \leq 4$ (1 + 3 + 30 classes; artifact), while $\tilde{H}_* \neq 0$ throughout (a nonzero H_3 in every case). For 24 of the 30 classes at $n = 4$ the inclusion-induced q_* already fails to be surjective on $\mathbb{F}_2$ -homology, which settles the factorization through the sieve there unconditionally.

Proposition 14.9 (Failure of non-domination at $n = 4$). *Part (i) of Hypothesis 14.4 is false in general. Among the six remaining classes at $n = 4$, three (of orders 8, 8, 24) have reflection-locus incidence posets whose order complexes are trees (5, 5, and 11 loci with no chains of length three; artifact), so $\text{el}(\Sigma)$ is contractible; by the collage formula (Theorem 10.1) the comma collapses onto BN and the sieve inclusion induces an equivalence $\text{el}(\Sigma/N) \xrightarrow{\simeq} \text{el}(\square^4/N)$. In particular $\text{el}(\square^4/N)$ is a homotopy retract of $\text{el}(\Sigma/N)$ there — consistently, the vertex Čech model gives $H_*(\text{el}(\Sigma/N)) \cong H_*(\text{el}(\square^4/N))$ with q_* an isomorphism in these three cases. The other three classes (orders 16, 32, 192) have incidence complexes with $\mathbb{F}_2$ -Betti numbers (1, 1, 0), (1, 5, 0), and (1, 0, 15); the last explains the excess $H_2(\text{el}(\Sigma/N)) \neq 0$ over $H_2(\text{el}(\square^4/N)) = 0$ exactly; at this stage of the development part (i) was undecided for these three — the dichotomy of Section 15 below settles all of them.*

Remark 14.10 (The governing invariant). Proposition 14.9 recasts the problem: what decides the sieve factorization at a reflection-generated mixed N is the homotopy type of the *incidence complex of the reflection arrangement* of N . At $n \leq 3$ every base has a noncontractible incidence complex (two points, four points, and a graph with $b_1 = 3$), which is why Hypothesis 14.4(i) is a theorem there (Proposition 15.11); at $n = 4$ tree arrangements appear for the first time and the test side no longer separates the quotient

from its sieve. Whether the identity factors through Σ/N up to *type* homotopy is then a separate, positive-decidable question, settled for two of the three tree classes by the following proposition.

Proposition 14.11 (Sieve factorization at tree arrangements). *On the Boolean site, for two of the three tree classes at $n = 4$ (the two of order 8) the identity of $\square^4/N$ factors through Σ/N up to type homotopy: there is an N -equivariant vertex map $r: \square^4 \rightarrow \square^4$ whose generic cell is stabilized (its image lies in a single plane stratum), and the pair (id, r) is a strict cylinder homotopy $(\square^4/N) \times \square^1 \rightarrow \square^4/N$ from the identity to $j \circ g$, where g is the class of r in $\text{Hom}(\square^4/N, \Sigma/N)$. Consequently the identity lifts along $\bar{\Phi}_N$ up to type homotopy, and $\bar{\Phi}_N$ is not a separating witness at these two classes. For the third tree class (order 24) no such one-step witness exists: an exhaustive search over all twisted-equivariance shapes $r \circ \sigma_h = \sigma_{c^{-1}hc} \circ r$ ($c \in N$) and all equivariant choices of values finds none (artifact).*

Proof. A one-step homotopy is a map $(\square^4/N) \times \square^1 \rightarrow \square^4/N$, i.e. a vertex function on $\{0, 1\}^5$ whose two slices share the same deck element for every $h \in N$; since the identity slice is a bijection its deck element is forced, so the second slice is exactly a c -twisted equivariant map for some $c \in N$, and it lands in the sieve iff its generic cell is stabilized. The cylinder $X \times \square^1 \rightarrow X$ is a genuine cylinder object in the type structure (every object is cofibrant, and connections contract $\square^1$), so a strict witness is an unconditional type homotopy; the search is exact because on the Boolean site cells are arbitrary vertex functions. The witnesses (and the exhaustion for the third class) are in the artifact; for one order-8 class there are 256 witnesses, for the other 32768. Lifting along $\bar{\Phi}_N$: the collapsed end includes Σ/N into J_N/N , and $\bar{\Phi}_N$ restricts to j there, so the end inclusion composed with g is the required lift. $\square$

Theorem 14.12 (Dimension reduction at tree arrangements). *For both order-8 tree classes, $\square^4/N$ is homotopy equivalent, simultaneously in both structures, to a lower-dimensional isotropy quotient by a mixed reflection-generated group: for the first class to $\square^2/K$ (the Klein quotient itself), for the second to $\square^3/R$ with R an order-4 mixed reflection-generated block. Consequently both classes are separation classes — the criterion’s law “SEP $\iff$ N mixed” survives at them — with the separating map obtained by transporting the lower-dimensional witness across the equivalence.*

Proof. In the witness census (artifact), every witness for the first class, and 8192 witnesses for the second, have image equal to the vertex set of a single stratum P whose setwise stabilizer is all of N , with $r|_P$ equal to an element of the residual group $R := \text{res}(P)$. Write $\iota: \square^d \rightarrow \square^4$ for the pattern embedding of P ($d = 2, 3$ respectively). Since the image of r lies in P and on the Boolean site vertex functions are maps, $r = \iota \circ \bar{r}$ with $\bar{r}: \square^d \rightarrow \square^d$, and equivariance of r makes $\bar{r}$ intertwine the actions along $N \twoheadrightarrow R$; hence $\bar{r}$ and ι descend to $\bar{g}: \square^4/N \rightarrow \square^d/R$ and $\bar{j}: \square^d/R \rightarrow \square^4/N$. Then $\bar{g} \circ \bar{j} = [r|_P] = \text{id}$

strictly ($r|_P \in R$), and $\bar{j} \circ \bar{g} = [r] \simeq \text{id}$ by the witness cylinder — a homotopy valid in both structures. So $\square^4/N \simeq \square^d/R$ in both. The residual R is mixed (no fixed vertex on P) and generated by reflections (artifact census), so $\square^d/R$ is a separation class by the $n \leq 3$ criterion; a homotopy equivalence transports the attached map $\bar{\Phi}_R \in W_{\text{test}} \setminus W_{\text{type}}$ along it, using two-out-of-three in each structure. $\square$

Remark 14.13 (The recursion). Theorem 14.12 corrects the expectation raised by Proposition 14.11: the death of $\bar{\Phi}_N$ at tree arrangements signals not agreement but *recursion* — the quotient is equivalent to the isotropy quotient of an invariant stratum by its residual, and the criterion’s law survives with a refined witness dictionary: at a noncontractible arrangement, $\bar{\Phi}_N$ itself separates; at a tree with an equivariant stabilized retraction, dimension reduction hands the class to a lower-dimensional mixed reflection-generated residual and the law recurses. For the order-24 tree class — test-side reduced to its sieve, but with no one-step type-side retraction — the question is whether the identity factors through the sieve beyond one step; the dichotomy of Section 15 answers no (the class is rigid), completing the dictionary.

15. THE SIEVE-LANDING DICHOTOMY

The one-step analysis behind Proposition 14.11 globalizes into a complete decision procedure. Throughout this section the site is Boolean and N is reflection-generated mixed.

The codiscrete setting. The recursion of this section leaves the cube quotients and lands in a slightly larger class of objects, which we fix once. For a finite set T , let $E(T) := \text{cosk}_0(T)$ be the *codiscrete object* on T : its $[k]$ -cells are all functions $\{0, 1\}^k \rightarrow T$, so that $\square^n = E(\{0, 1\}^n)$ on the Boolean site. An *instance* is a pair (T, M) with $M \leq \text{Sym}(T)$ fixed-point-free and generated by its fixed-point-having elements; the associated object is $E(T)/M$, and the cube quotients $\square^n/N$ of the criterion are the instances $(\{0, 1\}^n, N)$. Four standing facts make every argument below apply to instances verbatim: (i) maps from any presheaf into a codiscrete object are exactly vertex functions, so stage-zero maps $E(T)/M \rightarrow E(T')/M'$ are classes of vertex functions with the deck condition; (ii) $\text{el}(E(T))$ is contractible (the Čech nerve of T), $E(T) \times E_M$ is contractible and free, and the proofs of Theorem 14.6 and of the collage formula (Theorem 10.1) apply word for word — in particular $\text{el}(E(T)/M)$ is simply connected; (iii) the stabilized sieve consists of the cells valued in some $\text{Fix}(s)$, $s \neq 1$, whose loci and intersections are again codiscrete, hence contractible; (iv) the witness machine J_M over $E(T)$ and the sector machinery of Theorem 14.3 use only the coequalizer presentation of the replacement, connections and diagonals on the site, the contractibility and freeness of the body, and the fact that vertex stabilizers consist of fixed-point-having elements — all available for instances. We use

Lemma 15.1 and the trichotomy for instances throughout: their proofs are the same sector and first-exit inductions with $\square^n$ replaced by $E(T)$.

For an instance — we write it as N acting on the vertex set for uniformity with the cube case — call $F: \{0, 1\}^n \rightarrow \{0, 1\}^n$ *c-twisted* ($c \in N$) if $F \circ \sigma_h = \sigma_{c^{-1}hc} \circ F$ for all $h \in N$, and *twisted* if it is *c-twisted* for some c ; write $D_F(h) := \{d \in N : F \circ \sigma_h = \sigma_d \circ F\}$ for the deck sets.

Lemma 15.1 (Stage-zero visibility). *On the Boolean site, every class in $[\square^n/N, R(\square^n/N)]$ is represented by a strict cell (a vertex function with deck condition), and two strict classes are homotopic iff they are connected by a chain of strict cylinder homotopies $(\square^n/N) \times \square^1 \rightarrow \square^n/N$.*

Proof. This is the sector and first-exit machinery of Theorem 14.3 applied to $\square^n/N$ itself (decomposed along Σ/N and its free complement) and, for homotopies, to the same decomposition of $(\square^n/N) \times \square^1$: every replacement-level class reduces stagewise to a stage-zero class — the body part by the covering classification (free action on the cover, Lemma 8.3), the sieve part by induction on dimension — and on the Boolean site stage-zero cells are exactly vertex functions, and stage-zero cylinders exactly pairs of slices with common deck elements. $\square$

Lemma 15.2 (Freedom equals stabilization). *Let F be twisted. Then some deck set $D_F(h)$ has more than one element iff F is a stabilized cell (iff its class lands in Σ/N). Moreover: one-step homotopies from the identity class are exactly the twisted classes; every twisted class is one-step homotopic to the identity; and if no twisted map is stabilized, a one-step homotopy from a twisted class leads only to twisted classes.*

Proof. $d_1, d_2 \in D_F(h)$ iff $\sigma_{d_1^{-1}d_2} \circ F = F$, i.e. iff F is stabilized by $d_1^{-1}d_2$; and a level- n cell lies in Σ iff it is stabilized. A one-step cylinder from the identity has slices $(\sigma_a, \sigma_b \circ r)$ sharing deck elements; since σ_a is a bijection its deck element at h is forced to be aha^{-1} , which makes r $(a^{-1}b)$ -twisted; conversely, if F is c -twisted then the pair $(\sigma_a, \sigma_{ac}F)$ shares the deck elements aha^{-1} , so F is one-step homotopic to the identity. For the last claim: if all deck sets of all twisted maps are singletons, a shared-deck cylinder out of a translate of a twisted map forces the other slice to satisfy a full twisted equation (with the twist adjusted by the translations). $\square$

Theorem 15.3 (Sieve-landing dichotomy). *Exactly one of the following holds for each reflection-generated mixed N on the Boolean site.*

- (R) *Some twisted map is stabilized. Then the identity of $\square^n/N$ factors through Σ/N up to type homotopy by a single strict cylinder, and $\bar{\Phi}_N$ is not a separating witness (reduction).*
- (S) *No twisted map is stabilized. Then the strict homotopy component of the identity is exactly the set of twisted classes, which contains no constant class, no class landing in Σ/N , and no strictly invariant class. Consequently all three cases of the trichotomy (Theorem 14.3)*

are excluded for the identity and $\bar{\Phi}_N \in W_{\text{test}} \setminus W_{\text{type}}$ — no cohomological input is needed.

Proof. (R) is Proposition 14.11's argument. (S): by Lemma 15.2 the component of the identity under strict cylinder chains consists of twisted classes, and by Lemma 15.1 this computes the type homotopy component. A constant const_w is twisted only if $\sigma_{c^{-1}hc}$ fixes w for all h , i.e. only if N fixes a vertex — excluded for mixed N ; a class lands in Σ/N only if its cell is stabilized — excluded in case (S); a strictly invariant F that is also c -twisted satisfies $\sigma_{c^{-1}hc}F = F$ for all h , making it stabilized — excluded again. These are exactly the $\bar{\Phi}_N$ -images of the trichotomy's cases (b), (c), and (a) respectively: in particular the constant case is excluded combinatorially, with no cohomological input. $\square$

Theorem 15.4 (The complete classification for $n \leq 4$). *Every reflection-generated mixed class with $n \leq 4$ ($1 + 3 + 30$ classes) is decided, in closed form, by the dichotomy: the two order-8 tree classes of Theorem 14.12 are of type (R) and are separation classes by dimension reduction; the other 32 are of type (S), so $\bar{\Phi}_N$ separates (the vertex Čech computation $\tilde{H}_3 \neq 0$ in every case is consistency data, not an input). Hence*

$$\text{SEP}(H) \iff \langle\langle \text{Ref}(H) \rangle\rangle \text{ is mixed}$$

holds for every $H \leq B_n$ with $n \leq 4$, unconditionally: Hypothesis 14.4 is eliminated. (Existence of a stabilized twisted map is decided orbitwise: a stabilized c -twisted map stabilized by s exists iff for every vertex orbit representative v some vertex is fixed by $c^{-1}\text{Stab}(v)c$ and by every N -conjugate of s ; artifact.)

Proof. The census is the artifact's closed-form computation. For type (R), Theorem 14.12; for type (S), Theorem 15.3; for a general mixed H the attached witness descends along the free H/N -quotient as in the criterion theorem below, and for N non-mixed the agreement half applies. $\square$

Theorem 15.5 (Structure of reductions). *In case (R), let F be a stabilized twisted map and $S \leq N$ its stabilizer. Then:*

- (1) S is normalized by N , and the image of F lies in $\text{Fix}(S)$, an N -invariant locus.
- (2) Some power F^k restricts to a bijection of $T := \text{im } F^k$, an N -invariant vertex subset of $\text{Fix}(S)$, and $\text{id} \simeq [F^k]$ by a single cylinder.
- (3) $\square^n/N$ is homotopy equivalent, simultaneously in both structures, to $E(T)/N$ — the quotient of the codiscrete object on T (levelwise all functions into T), a subobject of Σ/N (Figure 5).
- (4) N acts on T with no fixed point, through its image $M \leq \text{Sym}(T)$; in particular $|T| \geq 2$.

Proof. (1) From $F\sigma_h = \sigma_{c^{-1}hc}F$ we get $\sigma_{c^{-1}hc}^{-1}F = F\sigma_h^{-1}$, so for $s \in S$: $\sigma_{(c^{-1}hc)s(c^{-1}hc)^{-1}}F = \sigma_{c^{-1}hc}\sigma_s F\sigma_h^{-1} = \sigma_{c^{-1}hc}F\sigma_h^{-1} = F$; as $h \mapsto c^{-1}hc$ is

a bijection of N , every N -conjugate of s stabilizes F . $\text{im } F \subseteq \text{Fix}(S)$ is the definition of stabilization, and $\sigma_h \text{Fix}(S) = \text{Fix}(hSh^{-1}) = \text{Fix}(S)$. (2) By induction F^k is c^k -twisted and S -stabilized; by finiteness some power restricts to a bijection of its image, and $\sigma_g T = F^k(\sigma_{c^k g c^{-k}} V) = T$; the homotopy is Lemma 15.2. (3) On the Boolean site cells are vertex functions, so the cells valued in T form the codiscrete subobject $E(T) \subseteq \square^n$, contained in Σ because the nontrivial S fixes T pointwise. F^k factors as $\iota \circ \bar{g}_0$, twisted equivariance descends to $\bar{g}: \square^n/N \rightarrow E(T)/N$ and $\bar{j}: E(T)/N \rightarrow \square^n/N$, and $\bar{j}\bar{g} = [F^k] \simeq \text{id}$ while $\bar{g}\bar{j} = [F^k|_{E(T)}]$ has strictly invertible power ($F^k|_T$ is a bijection of finite order), so both composites are isomorphisms in each homotopy category and $\bar{g}$ is an equivalence. (4) A fixed point of N on T would be an N -fixed vertex of $\square^n$, contradicting mixedness; $|T| = 1$ would be such a fixed point. $\square$

$$\begin{array}{ccccc}
 & & \bar{g}\bar{j}=[F^k|_T] \in \text{Aut} & & \\
 & & \downarrow & & \\
 \square^n/N & \xrightarrow{\bar{g}=[F^k]} & E(T)/N & \xrightarrow{\bar{j}} & \square^n/N \\
 & \searrow & & \nearrow & \\
 & & \text{id} \simeq \bar{j}\bar{g} & &
 \end{array}$$

FIGURE 5. Dimension reduction at a case-(R) class (Theorem 15.5): the stabilized twisted map F^k factors through the codiscrete object on its eventual image T , the composite on $E(T)/N$ is a strict finite-order automorphism, and the outer composite is a one-step cylinder homotopy — an equivalence in both structures at once.

Corollary 15.6 (The recursion is well-founded). *The dichotomy theory — twisted maps, deck sets, the sieve of stabilized cells, Lemmas 15.1 and 15.2, and Theorem 15.3 — uses only that cells on the Boolean site are vertex functions, so it applies verbatim to codiscrete quotients $E(T)/M$. By Theorem 15.5, each case-(R) step replaces (T, M) by a strictly smaller fixed-point-free instance, so the recursion terminates at a case-(S) leaf with $|T| \geq 2$, and the leaf’s separating map transports up the chain of equivalences by two-out-of-three. At $n \leq 4$ both reduction chains have length one and land on full loci — the Klein quotient and an order-4 mixed block. The leaf instances are again “reflection-generated mixed” in the $E(T)$ -sense by the following transport lemma, so the trichotomy machinery applies to them.*

Lemma 15.7 (Reflection transport). *In every reduction, the image M of N in $\text{Sym}(T)$ is generated by its fixed-point-having elements. More precisely: if N is generated by a conjugation-closed set R of elements each having a fixed point (for $N \leq B_n$, its reflections; Lemma 11.1), then every element of R fixes a point of T .*

Proof. Let $\tilde{\rho} \in R$ and set $\rho := c^k \tilde{\rho} c^{-k} \in R$ (conjugation-closed), with fixed point v . Twisted equivariance of F^k gives $F^k \circ \sigma_\rho = \sigma_{c^{-k} \rho c^k} \circ F^k = \sigma_{\tilde{\rho}} \circ F^k$, so $F^k(v) = F^k(\rho v) = \sigma_{\tilde{\rho}}(F^k(v))$: the element $\tilde{\rho}$ fixes the point $F^k(v) \in T$. Since $N = \langle R \rangle$, the image M is generated by the images of R , each with a fixed point; and the fixed-point-having generating set of a leaf is again conjugation-closed, so the lemma applies at every level of the recursion. $\square$

Theorem 15.8 (The criterion at every n). *On the Boolean site, for every finite set T and every $M \leq \text{Sym}(T)$ generated by its fixed-point-having elements:*

$$\text{SEP}(E(T)/M) \iff M \text{ has no fixed point on } T,$$

and consequently, for every n and every $H \leq B_n$,

$$\text{SEP}(H) \iff \langle\langle \text{Refl}(H) \rangle\rangle \text{ is mixed,}$$

with no hypotheses beyond the standing machinery (Lemma 15.1 and the trichotomy for codiscrete bases). Figure 3 displays the recursion.

Proof. Induction on $|T|$. If M fixes t_0 , the constant at t_0 is twisted (its deck elements can all be taken trivial), so the identity is one-step homotopic to a constant and $E(T)/M$ is contractible in both structures: agreement. If M is fixed-point-free, apply the dichotomy (Theorem 15.3): in case (S), $\bar{\Phi}_M$ separates; in case (R), Theorem 15.5 gives an equivalence, in both structures, to a strictly smaller instance $E(T')/M'$ which is fixed-point-free (Theorem 15.5(4)) and generated by fixed-point-having elements (Lemma 15.7); the inductive witness transports across the equivalence by two-out-of-three. For $H \leq B_n$: if $N := \langle\langle \text{Refl}(H) \rangle\rangle$ is non-mixed the agreement half applies; if mixed, N is generated by its reflections — fixed-point-having on the vertex set — so the first statement attaches a witness to $\square^n/N = E(\{0,1\}^n)/N$ restricted along the faithful image, and it descends along the free H/N -quotient as in Theorem 16.1. $\square$

The remainder of this section documents, for the record, the conditional route that preceded the dichotomy — the following theorem and Proposition 15.11 are superseded by Theorem 15.8, but the low-dimensional $\mathbb{F}_2$ -computations of Proposition 15.11 remain of independent interest for the arrangement dichotomy of the outlook.

Theorem 15.9 (Separation at reflection-generated groups; superseded). *Under Hypothesis 14.4, $\bar{\Phi}_N \in \mathcal{W}_{\text{test}} \setminus \mathcal{W}_{\text{type}}$ for every reflection-generated mixed N .*

Proof. Test: Proposition 14.2. Type: if $\bar{\Phi}_N$ were a type weak equivalence, the identity of $\square^n/N$ would lift along it up to homotopy, hence by the trichotomy of Theorem 14.3 the identity would be homotopic to a constant, to a class factoring through Σ/N , or to a strictly invariant endomorphism class. The first two are excluded by Hypothesis 14.4(i) after localizing to

the test side (contractibility being the empty-sieve case of domination), and the third by Hypothesis 14.4(ii). $\square$

Lemma 15.10 (Degree-one generation). *If Y is a homotopy retract of X , with section s and retraction r , then $s^*: H^*(X; \mathbb{F}_2) \rightarrow H^*(Y; \mathbb{F}_2)$ is a surjective ring homomorphism. In particular, if $H^*(X; \mathbb{F}_2)$ is generated as a ring in degrees ≤ 1 , then so is $H^*(Y; \mathbb{F}_2)$; if moreover $H^1(Y; \mathbb{F}_2) = 0$, then $\tilde{H}^*(Y; \mathbb{F}_2) = 0$.*

Proof. $(r \circ s)^* = s^* \circ r^* = \text{id}$, so s^* is surjective, and it is a ring map because it is induced by a map of spaces. A surjective ring map carries a generating set onto a generating set; if the target has no nonzero elements in degree one, the images of the degree- ≤ 1 generators lie in H^0 , so the reduced part of the target vanishes. $\square$

Proposition 15.11 (Hypothesis 14.4 in low dimensions). *Hypothesis 14.4 holds for every base occurring in the $n \leq 3$ classification.*

Proof. The bases are the Klein block groups, the A_4 -type group, and the nested order-24 group. (i) is settled uniformly by Lemma 15.10. Every component of the stabilized sieve quotient is a point or a $B\mathbb{Z}/2$: for the Klein group, $\text{el}(\Sigma/K) \simeq B\mathbb{Z}/2 \sqcup B\mathbb{Z}/2$ (two isolated reflection loci); for the A_4 -type group, $\Sigma/N \cong \square^1$, so $\text{el}(\Sigma/N) \simeq *$; for the nested group, the plane stratum contributes $\text{el}(\square^2/R) \simeq B\mathbb{Z}/2$ (the residual R is a free reversal), the line stratum a point, and the gluing is along a point, so $\text{el}(\Sigma/N) \simeq B\mathbb{Z}/2$. In each case $H^*(\cdot; \mathbb{F}_2)$ is generated in degrees ≤ 1 (as $\mathbb{F}_2[t]$, or trivially), and a connected retract of a disjoint union retracts through one component. On the other side, the collage of Theorem 10.1 computes, by exact $\mathbb{F}_2$ -cochain models (normalized bar complexes and iterated homotopy pushouts; artifact):

$$H^*(\text{el}(\square^2/K)) = (1, 0, 0, 1, 2, 3, \dots), \quad H^*(\text{el}(\square^3/N_{A_4})) = (1, 0, 1, 2, 1), \quad H^*(\text{el}(\square^3/N_{24})) = (1, 0, 0, 1, \dots)$$

the first being the join $B\mathbb{Z}/2 \star B\mathbb{Z}/2$, the second equal to $H^*(BN_{A_4})$ (the odd comma $B(\mathbb{Z}/3)$ is $\mathbb{F}_2$ -acyclic), the third assembled from the graph-of-groups comma $B(\mathbb{Z}/2 \times \mathbb{Z}/2) \leftarrow B(\mathbb{Z}/2) \rightarrow B(S_3)$. In all three, $H^1 = 0$ while $\tilde{H}^* \neq 0$ (in degree 3, 2, 3 respectively), so by Lemma 15.10 none is a homotopy retract of its stabilized sieve quotient, nor contractible. (ii) is the stage-zero verification of the preceding remark (artifact). $\square$

16. THE CRITERION THEOREM

Theorem 16.1 (The criterion). *Let $H \leq B_n$ be mixed and $N := \langle\langle \text{Refl}(H) \rangle\rangle$. (1) On any reversal site, if N has a common fixed cell then $\square^n/H \simeq B(H/N)$ in both structures. (2) If N is mixed then an explicit test weak equivalence that is not a type weak equivalence is attached to H : on the Boolean site for every n , and on the De Morgan and Kleene sites for $n \leq 3$.*

Proof. (1) is Theorem 13.1. (2): N is reflection-generated (the closure is generated by the H -conjugates of reflections of H , which are reflections) and mixed, so Theorem 15.8 attaches a separating map to N on the Boolean site; H/N contains no reflections, acts freely on $\square^n/N$ and on the witness objects, and the covering descent theorem (Theorem 8.4) descends the separation to H along the free quotient. For $n \leq 3$ the same conclusion transfers to the De Morgan and Kleene sites by Part 2, and the criterion reproduces the classification of [9] on all 98 subgroups with zero mismatches (Part 4). $\square$

17. OUTLOOK

The sieve-landing dichotomy (Theorem 15.3) replaces Hypothesis 14.4 entirely, needs no cohomological input, and — with the reflection-transport lemma — yields the criterion at every n (Theorem 15.8), with $n \leq 4$ decided in closed form (Theorem 15.4); the standing machinery is stated at codiscrete generality in Section 15, its proofs being the sector and first-exit inductions of Theorem 14.3 with the cube replaced by $E(T)$. What remains open beyond the comparison program itself is the converse of the Sylow cone below, and the Dedekind-side questions of the phase diagram.

Independently of the criterion, the cohomology of the orbit spaces $E(T)/M$ — simply connected when M is generated by fixed-point-having elements (Theorem 14.6), computed by the collage formula from the fixed-point arrangement — exhibits a dichotomy of its own. One direction is a theorem:

Proposition 17.1 (Sylow cone). *If a Sylow 2-subgroup P of M has a fixed point $t_0 \in T$, then $E(T)/M$ is $\mathbb{F}_2$ -acyclic.*

Proof. The extra degeneracy $s_P(x_0, \dots, x_k) := (t_0, x_0, \dots, x_k)$ is P -equivariant (t_0 is P -fixed) and contracts the augmented chain complex $\tilde{C}_*(E(T); \mathbb{F}_2)$ over $\mathbb{F}_2 P$. The relative trace $s := \sum_{mP \in M/P} \sigma_m s_P \sigma_m^{-1}$ is well defined and M -equivariant, and $ds + sd = [M:P] \cdot \text{id} = \text{id}$ over $\mathbb{F}_2$; applying $(-)_M$ gives a contraction of $\tilde{C}_*(E(T)/M)$. $\square$

Hence nonvanishing forces the Sylow 2-subgroups to be fixed-point-free, and the computations (twelve cases, artifact) are consistent with the converse for the criterion-relevant class:

- (D) for M generated by its fixed-point-having elements, $\tilde{H}_*(E(T)/M; \mathbb{F}_2) \neq 0$ iff the Sylow 2-subgroups of M have no fixed point in T .

Outside that class the nonvanishing can be purely fundamental-group-driven ($\mathbb{Z}/4$ acting on a regular orbit plus a 2-orbit has $\tilde{H}_* = (1, 1, 1, \dots)$ from $\pi_1 = \mathbb{Z}/2$), while inside it, S_3 on three points is fixed-point-free with Sylow fixed points and $\mathbb{F}_2$ -acyclic orbit space (though not $\mathbb{F}_3$ -acyclic) — yet still a *separation* instance of the criterion, its twisted maps forced onto the full vertex set: the combinatorial dichotomy is strictly sharper than $\mathbb{F}_2$ -cohomology. Every mixed reflection-generated cube class with $n \leq 4$ contains even a fixed-point-free elementary abelian 2-subgroup (artifact), so

the cube world lies on the conjecturally nonvanishing side of (D) throughout — consistently, $\tilde{H}_3 \neq 0$ is observed in every case — and (D) is the promised bridge from the model-structure landscape to the mod-2 homotopy theory of finite group actions on finite sets, with $E(T)$ the classifying space of the family of subgroups with fixed points. Beyond it, two directions. The Dedekind question — whether the connective site without reversal presents spaces — is untouchable by isotropy witnesses (Theorem 3.1) and remains the outstanding open case of the phase diagram. And the criterion suggests a dictionary with reflection arrangements: the fixed loci of $\text{Refl}(H)$ form an arrangement of subcubes, and the two homotopy theories disagree exactly when the arrangement’s symmetry cannot be disentangled from a fixed stratum; a Coxeter-theoretic form of the criterion is a natural sequel.

Part 4. Machine verification

18. THE VERIFICATION SUITE

Every finite combinatorial claim of this paper is verified by exhaustive computation, in the scripts of the artifact repository of the series,

<https://github.com/r-shibusawa/cubical-strict-j>

releases v1.16.0–v1.19.0 (version-pinned, with Zenodo DOIs on the release pages; the release accompanying this paper is pinned at submission time). The main entries:

- **kleene.site.py, boolean.site.py**: the Kleene (unmixed-poset) and Boolean machineries; parity lemmas at the unmixed witness points; stage-zero censuses of the Klein machines (essential classes 0/18/2 across DM/KL/BA); the collage factorization $F(d, a, t) \oplus F(d, \neg a, t) = t$.
- **boolean.delta.py, kleene.delta.py, boolean.certificates3.py, kleene.certificates3.py**: the exact certificate systems (Boolean: parity-exact; Kleene: with the monotone-realizability filter) for the two-stratum family and the six beyond-two-stratum groups of dimension three; survivor statistics of Table 2.
- **boolean.fate.py**: interior realizations of consistent twists, exact slide-null decisions (positives verified by reconstruction), collage images — the data behind the rigidity discriminant, 12/12.
- **discriminant.check.py**: the discriminant ($\varepsilon \circ \delta$) against the fate data, and the reflection-generation characterization on fourteen groups across $n = 2, 3, 4$ — all matching.
- **boolean.certificates4.py, degenerate.search.py**: the five dimension-4 primitive groups $V_{\text{even}}(4) \rtimes P$ (all certified) and the five degenerate groups with no one-step equivariant consensus pattern.
- **separation.criterion.py**: the criterion against the family recursion of [9] on all 98 subgroups of B_3 : 78 agreement / 20 separation,

zero mismatches; the reflection closure is generated by its own reflections in every case; every reflection fixes a vertex.

- `isotropy_poset.py`, `el_betti.py`, `el_betti_a4.py`, `el_betti_nested.py`: the incidence posets of isotropy loci with setwise stabilizers and residual actions; the exact $\mathbb{F}_2$ -cochain engine for the collage formula (normalized bar complexes, induced cochain maps, homotopy pushout models with tracked cocycle representatives); the computations of Proposition 15.11, validated against the closed-form join $B\mathbb{Z}/2 \star B\mathbb{Z}/2$ and against $H^*(BN_{A_4}; \mathbb{F}_2)$.
- `edom_check.py`, `realizability_check.py`: stage-zero verification of Hypothesis 14.4(ii) for the $n \leq 3$ bases (the hypothesis is superseded by the dichotomy in the body; the check documents the historical route), and the unified realizability theorem (25/25 across the three sites).
- `frontier_census.py`, `frontier_census2.py`, `q4_check.py`: the six $n = 4$ classes left undecided by q_* (all later settled by the census) — incidence complexes (trees, S^1 , $(1, 5, 0)$, $(1, 0, 15)$), and q_* -surjectivity through degree 4.
- `tree_factor_check.py`, `nontree_factor_check.py`, `tree_witness_analysis.py`, `twostep_factor_check.py`, `rigidity_census.py`: the sieve-landing dichotomy — one-step witnesses (256 and 32768 at the two reduction classes, none elsewhere), witness structure (single-stratum bijective images for all witnesses of the first class and 8192 of the second; residuals K and an order-4 mixed block), deck-freedom counts, and the closed-form census of Theorem 15.4 over all 34 classes.
- `reduction_structure.py`: all four claims of Theorem 15.5 and the conclusion of Lemma 15.7, verified on every witness of both reductions (256 + 32768).
- `etm_library.py`, `etm_decisive.py`, `etm_decisive2.py`: the twelve M -set computations behind the arrangement dichotomy — the $\mathbb{F}_2$ - and $\mathbb{F}_3$ -acyclicity of S_3 on three points, the decisive $(\mathbb{Z}/2)^2$ on three 2-orbits and $\mathbb{Z}/4$ on a regular orbit plus a 2-orbit, and the fixed-point-free elementary abelian 2-subgroups in all 34 cube classes.

On the Boolean site every decision procedure above is *exact*: truth tables are unconstrained, so a consistent parity assignment is a cell, and positives are re-verified by reconstruction. On the Kleene site exactness is restored by the monotone-realizability filter over the unmixed poset. Several of the paper’s theorems were found as empirical laws in this data and then proved: the rigidity discriminant, the realizability divide, the reflection-generation characterization, and the criterion itself.

Data availability. [Artifact, Zenodo DOI at release time.]

Use of AI tools. An AI assistant was used in the preparation of this manuscript and its verification scripts; the author takes full responsibility for the content.

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